# SkillMerge Supplement: Conditional Safety, Counterfamilies and Reproduction Ledger

Research Forge · 19 October 2026 · Layout v21 · Research content v21

New v21 theorem and deterministic reanalysis are in S15-S24. S1-S14 remain the exact v20 explanations; inherited.paper remains the complete unchanged v19 scientific record. All exact v20 document bytes are archived in preserved_v20/. No independent skill experiment, native benchmark or model call is added.

**Abstract**

Detailed assumptions, derivations, full comparison contracts, negative evidence and reproducibility notes accompany the compact main paper. The full previous scientific record is retained without changing its reported numbers or evidence scope. New finite computations reuse existing exposed evidence; no model invocation or independent empirical sample is introduced. New v21 sections distinguish these calculations from preserved v20 explanations and the v19 scientific record.

## 1 S1. Scope and exact reading map

The compact main text answers one operational question: what changes when common evidence IDs are promoted relative to a fixed round-robin order? This supplement makes the claim conditions inspectable. It separates four objects: a finite saved-bank result; a mathematical proposition under explicit assumptions; an authored example used to explain or refute a stronger proposition; and an unexecuted future behavioral protocol. None is silently substituted for another.

S2–S7 develop the output and reference contracts, proofs, boundary cases, and witnesses. S8–S10 explain denominators, exact aggregate results, comparison strengths, and selection rights. S11–S12 preserve reproducibility bindings and the verifier correction. S13–S14 specify what additional evidence would permit a behavioral decision. No new figure is inserted in these added sections: the complete inherited manuscript retains its original figures and historical discussions once, in the later archival portion.

The inherited paper object is a semantic JSON copy of projects/research_advancement_v19_20261019/skillmerge/paper.json, whose original byte hash is recorded in inherited.source_sha256. Its 21 reference objects, numbers, URLs, and claim_scope strings remain unchanged. A byte hash identifies which source was copied; it does not certify the source's semantic truth, paper novelty, or an external review. The Korean explainer preserves its complete v19 text after a new reading guide. STUDY_MAP.json provides machine-readable exact-heading links.

| Main heading | Supplement destinations | Korean guide heading |
|---|---|---|
| Introduction | S1. Scope and exact reading map; S14. Admission decisions and falsification ledger | 1. 일곱 절을 읽는 지도 |
| Related Work | S10. Strong baselines and restricted selection rights | 2. 그림과 정답 묶음을 먼저 구별하기 |
| Problem and Method | S2. Evidence contracts and boundary conventions; S3. Stable operators and information permissions; S4. Completion ranks and the single-window proof; S5. Removed IDs and an exact loss certificate; S6. Conditional cap-four theorem and its limits | 6. 수식의 기호와 증명을 읽는 순서 |
| Experimental Setup | S8. Paired denominators and weighting reversal; S11. Frozen execution and provenance ledger; S12. Failed verification and the corrected claim | 5. 손계산 3: 같은 자료의 가중치가 부호를 바꾸는 이유 |
| Results | S7. Saved and minimal failure witnesses; S9. Complete capacity results and endpoint tradeoffs | 3. 손계산 1: 공통 항목 하나가 밀어내는 것 |
| Limitations | S6. Conditional cap-four theorem and its limits; S13. Prospective independent-history protocol | 4. 손계산 2: 용량 4는 왜 무회귀이지 전승이 아닌가 |
| Conclusion | S14. Admission decisions and falsification ledger | 7. 혼동을 막는 자기점검과 다음 연구 |

Table 1: Exact compact-main headings and their expanded reading destinations. S numbers are supplement headings, not extra empirical studies.

## 2 S2. Evidence contracts and boundary conventions

Fix one evidence-bearing task and its finite sentence-ID domain. Its reference family consists of one or more nonempty sets. Several alternatives may support the same task; an output need contain one complete alternative, not the union of all alternatives. In the authored finite certificate, alternatives

may overlap. That broader certificate class is explicitly different from claiming that every such family occurs in the native dataset. No independent domain expert adjudicated reference completeness or correctness.

A and B are ordered lists without duplicate IDs. Empty input lists are admitted in boundary checks, but an empty input cannot be annotation-sufficient when all references are nonempty. A valid generated order likewise contains distinct IDs. For nonnegative integer k, its prefix contains the first min(k,length) entries. Cap zero yields the empty output. Capacities beyond the union size do not pad or invent IDs. An ID outside the output order has infinite rank; the infinity convention concerns absence, not a numeric rank inserted into the response.

Full containment S asks whether at least one reference is a subset of the entire output. First-three asks the same question after an additional prefix-three operation. Exact-annotated E asks whether the entire output set equals one listed alternative. An empty output scores zero for all three on evidence-bearing tasks. The original broader task audit also has insufficient-evidence tasks with a separate label/empty-evidence contract; those are not relabeled as members of the current 309-task pair bank.

These endpoints can disagree legitimately. Adding an unannotated ID preserves full containment but can destroy exact matching. Moving a required ID behind position three preserves the full set but can destroy first-three acceptance. An unannotated ID is not automatically semantically irrelevant, and an exact-match failure is not automatically an incorrect answer. Conversely, if the application truly requires a short ranked rationale, its first-three failure is operationally real even when the evidence still occurs later. Correct labels are supplied by the auditor throughout; label prediction, retrieval from a corpus, native aggregate F1, and model faithfulness are outside the executed contract.

$$S(X) = \mathbb{1}[\exists R \in \mathcal{R} : R \subseteq X] \tag{1}$$

$$F_3(L) = S(\mathrm{set}(L[1 : \min(3, |L|)])) \tag{2}$$

$$E(X) = \mathbb{1}[\exists R \in \mathcal{R} : X = R] \tag{3}$$

## 3 S3. Stable operators and information permissions

The complete round-robin order V is constructed before the output cap. Visit A's first entry, B's first entry, A's second entry, B's second entry, continuing through the longer list. Append an ID only on its first visit. Thus V contains each element of the union once. Define C as the intersection. P is the C-subsequence of V followed by the non-C subsequence of V. This is a stable partition, not sorting by ID value, reference membership, or an inferred semantic relevance score.

The policy input is exactly A, B, and k. Reference families are passed only to the scoring/auditing stage. The construction code does not receive reference labels, completed-reference ranks, the both-input-sufficient inclusion flag, or a held-out score. Completion-rank and removed-ID certificates are gold-aware explanations after construction; they are not exposed as routing features. A policy that used those certificates to select its output would have different information rights and could not inherit the label-blind comparison.

The earlier agreement-first rule obtained the relative order of common IDs from a left-priority construction. The current stable partition obtains both class orders from V. At capacity three, every common ID fits because each input has at most three IDs; set-based endpoint results match the saved priority audit. At capacities one and two, a prefix may contain only some common IDs, so the change in internal class order matters. These small-cap results are not described as an identical rerun of the previous constructor.

An interleaving, a common-item promotion, truncating an input before union construction, and truncating the completed union are different interventions. Only the last specified sequence is analyzed. An implementation that drops entries from A or B before deduplication, sorts common IDs numerically, consults a reference, or silently fills an empty intersection with oracle IDs changes the experiment. The reproducibility check must compare exact ordered outputs as well as their scores: tied scores alone cannot establish that the intended constructor was executed.

INPUT A, B, nonnegative integer k; initialize V=[], seen=empty set.
For index i from 0 to max(len(A),len(B))-1, visit A[i] then B[i] when present.
Append a visited ID to V only if it is not already in seen; then add it to seen.
Set C=set(A) intersect set(B).
Set P=[x in V with x in C] followed by [x in V with x not in C].
Return the prefix of V and the prefix of P of length min(k,len(V)).
Only after outputs are frozen, evaluate the original nonempty reference alternatives.

Algorithm 1: Deterministic label-blind constructor. References are not an argument. The scorer is a separate audit step.

## 4 S4. Completion ranks and the single-window proof

Fix any complete distinct-ID order L and nonempty alternative R. If every member of R appears in L, the smallest prefix containing R ends at its last required ID. Its completion rank is therefore the maximum of the member ranks. If at least one member is absent, no finite prefix completes R and that alternative's completion rank is infinity. The family is accepted as soon as its earliest-completing alternative is accepted, so take the minimum over alternatives. This establishes the formula in both the finite and absent-reference cases without assuming equal reference lengths or disjoint alternatives.

To prove the prefix identity in both directions, suppose k is at least the minimum completion rank. A minimizing finite alternative has every member ranked at most k and is a subset of the prefix, giving sufficiency. Conversely, if the prefix is sufficient, some alternative lies entirely inside it, so its maximum member rank is at most k and the minimum completion rank is no larger. If all alternatives have infinite completion rank, no finite prefix passes. The minimum is attained whenever it is finite because the recorded family is finite.

Let b and p be the baseline and promoted completion thresholds. If p is below b, the promoted output alone passes exactly for p <= k < b. Before p both fail; from b onward both pass. If b is below p, baseline alone passes exactly for b <= k < p. Equal thresholds give ties at every capacity, including the case where both are infinite. When one threshold is infinite the interval is unbounded above over unrestricted capacities; on the audited finite range it is intersected with zero through six. For two permutations of the same full union, reachability is shared, so either both thresholds are finite or both infinite.

This proves a single contiguous non-tie interval and a single polarity for each fixed pair under full containment. It does not prove a one-capacity interval. A manually authored teaching example uses A=(1,2,0), B=(4,5,0), but changes the reference family to the singleton alternative {0}. This is not the saved task's original annotation. V=(1,4,2,5,0) has threshold five; P=(0,1,4,2,5) has threshold one. Gains then span capacities one through four. The saved bank's observed width-one intervals are a property of that bank, not a theorem about stable promotion.

Exact matching lacks the same monotonicity: a prefix can equal an alternative and then stop equaling it after an extra ID is added. First-three is monotone only until its effective cap reaches three and then freezes. The rank formula can describe that frozen endpoint by replacing k with min(k,3), but increasing generated k beyond three cannot restore a reference whose earliest required rank exceeds three. Interpreting the rank threshold as a model confidence or a semantic correctness guarantee would add an unsupported assumption.

$$r_R(L) = \max_{x \in R} \operatorname{rank}_L(x) \tag{4}$$

$$r(L) = \min_{R \in \mathcal{R}} r_R(L) \tag{5}$$

$$S_k(L) = \mathbb{1}[k \geq r(L)] \tag{6}$$

$$S_k(P) - S_k(V) = \begin{cases} 1 & p \leq k < b \\ -1 & b \leq k < p \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

## 5 S5. Removed IDs and an exact loss certificate

Write Bk and Pk for the baseline and promoted prefix sets. A common ID x already in Bk has common-class rank at most its original V rank. Moving the entire common subsequence to the front gives it that common-class rank, never a later one. Thus every baseline-selected common ID remains selected. A noncommon ID retained by Pk has noncommon rank at most k minus the number of common IDs filling the promoted prefix; enough earlier noncommon positions therefore existed in the baseline prefix to contain it. Equivalently, stable promotion cannot add a previously unselected noncommon ID. These statements also hold when k exceeds the union size because both outputs then contain the full union.

Consequently, removed IDs D=Bk minus Pk are noncommon, and added IDs H=Pk minus Bk are common. The two output sets have equal size, so D and H have equal size. If a baseline-complete reference contains only common IDs, it survives. More generally, a baseline-complete reference survives exactly when it avoids D, because all of its remaining IDs were already in Bk and are retained. This gives a useful necessary risk condition without asserting that common IDs are true or false.

A full loss occurs if and only if at least one reference was complete in Bk, every such complete reference intersects D, and no reference is complete in Pk. The final condition matters: destroying all previously complete alternatives does not prevent a different alternative from becoming complete using the newly added common IDs. Checking only one preferred annotation would also be inadequate; the loss certificate must quantify over the original family. The saved output stores the relevant removals, additions, and alternatives so a reader can audit this distinction.

There is a separate universal boundary. If two equal-size output sets differ, choose an ID x present in the baseline and absent from the other output. Under the admissible singleton reference family {{x}}, the baseline passes and the other fails. Therefore, without restrictions on possible references, a same-size transformation that changes the set cannot guarantee full-containment no-regression for every family. This is a quantified annotation-contract statement, not an impossibility result for useful skill integration. It does not say every changed output loses on the actual task, and it does not license choosing a favorable artificial reference after seeing model behavior.

The universal singleton construction need not satisfy the both-input-sufficient restriction used in S6: the removed noncommon ID may occur in only one input. Its counterexample therefore cannot refute that conditional theorem. The order of quantifiers matters: a guarantee for every reference family is stronger than a guarantee only for families witnessed in both complete inputs.

$$D = B_k \setminus P_k, \quad H = P_k \setminus B_k \tag{8}$$

$$D \subseteq (A \cup B) \setminus C, \quad H \subseteq C \tag{9}$$

$$|D| = |H| \tag{10}$$

## 6 S6. Conditional cap-four theorem and its limits

The theorem assumes: A and B each contain at most three distinct IDs; each full input contains at least one complete nonempty reference alternative; the complete deduplicated union is formed before truncation; the common and noncommon classes preserve their relative V order; and the endpoint is full-output containment. The two inputs may contain different reference alternatives. This is an annotation condition on full stored lists, not an empirical Q gate on independently authored programs and not a guarantee that each shortened input is sufficient.

Partition by the common-set size c. If c=0, P=V exactly, so no baseline-success-to-promotion-failure transition is possible at any capacity. This establishes no-loss even if both prefixes fail. If c>=2, the union has at most 3+3-2=4 IDs. At k>=4 both outputs therefore contain the union, and the sufficient-input assumption implies success. If c=1, the union has at most five IDs. At k>=4 the promoted prefix retains the common ID and omits at most one union ID. An omitted noncommon ID belongs to only one input, so the other input is entirely retained; its complete reference alternative remains. These cases exhaust the input class.

The distinction between success and no-loss is indispensable. Take A=(0,1,2), B=(3,4,5), with alternatives {0,1,2} and {3,4,5}. Both full inputs are sufficient, but at capacity four both orders are (0,3,1,4) and neither alternative is complete. The implication 'baseline passes implies promotion passes' is true because the antecedent is false; unconditional promoted success is false. This is the condition that exposed the initial verifier's overstrong assertion. Likewise, the theorem allows gains: S4's authored common-singleton example has a capacity-four gain. The bank's all-tie result above three is not the theorem's conclusion.

The input-length restriction is substantive. In a manually derived illustration, let A=(1,2,3,0), B=(4,5,0), with alternatives {1,2,3} and {4,5}. The complete inputs are sufficient and the partition is stable, but one input has length four. V=(1,4,2,5,3,0) at cap four passes via {4,5}; P=(0,1,4,2,5,3) at cap four completes neither alternative. This is an explanatory counterexample outside the theorem's class, not an additional observed task or a rerun of the finite benchmark.

First-three and exact endpoints remain outside the guarantee. A generated fourth ID can complete a reference that first-three never reads, while exact matching may reject a sufficient superset. Truncating inputs before union formation also changes which alternatives can survive. Empty references are deliberately excluded: admitting an empty valid alternative would make full containment vacuously

pass, changing the meaning of qualification and all cap-zero results. Boundary handling is part of the proposition, not a detail to fix after seeing the score.

## 7 S7. Saved and minimal failure witnesses

The preserved actual diagnostic witness is scifact_v4_382. A=(1,2,0) contains alternative {1,2}; B=(4,5,0) contains alternative {4,5}. The shared ID is zero. V=(1,4,2,5,0) and P=(0,1,4,2,5). At cap three the baseline retains IDs one and two, whereas promotion removes two and adds zero. It thereby completes neither original alternative. The inputs, union, output count, references, and noncommon order are fixed; common-class priority identifies this particular transition.

At cap four, promoted (0,1,4,2) again includes {1,2}. Its first-three prefix remains (0,1,4), so generated length and evaluator length tell different stories. At cap five or six, both operators emit all five IDs and full containment passes; first-three still reads the same failing promoted prefix. No claim about the semantic irrelevance of sentence zero is needed for the set calculation. The two recorded losses in the bank are ordered pair views, not two independent trials of real skills.

The separate authored certificate enumerates ID universes zero through six, all distinct-ID ordered inputs of lengths zero through three whose union is that universe, and one/two-alternative families witnessed in both full inputs. It covers 3,046 input pairs and 122,571 families, including overlapping alternatives, at seven capacities. Lexicographic witness selection records the smallest union size and then total input length where losses occur. This is minimality within the declared exhaustive finite class and selection order, not a global minimal natural-language counterexample.

| Cap / union / total input | A ; B | Alternatives | Baseline ; promoted prefix |
|---|---|---|---|
| 1 / 3 / 4 | (0,1) ; (1,2) | {0} or {1,2} | [0] ; [1] |
| 2 / 4 / 5 | (0,1,2) ; (3,1) | {0,1,2} or {3} | (0,3) ; (1,0) |
| 3 / 5 / 6 | (0,1,2) ; (3,2,4) | {0,1} or {2,3,4} | (0,3,1) ; (2,0,3) |
| 0 and 4..6 | No loss in the declared class | Nonempty alternatives | No-loss, not unconditional success |

Table 2: Saved certificate witnesses for the both-input-sufficient class. These authored families are not new independent dataset observations.

## 8 S8. Paired denominators and weighting reversal

A pair-capacity transition is a matched comparison: score the baseline and promotion on the same ordered inputs, task references, endpoint, and cap. A gain is a zero-to-one change; a loss is one-to-zero; ties include both failure/failure and success/success. The pair-pooled contrast equals gains minus losses divided by all retained pairs. Dropping unchanged failures, empty outputs, or zero-common pairs changes the denominator and cannot be called the same effect. A separate qualified-history target would additionally need actual independent authoring and an observed Q gate, which are absent.

The full cohort contains 22,794 ordered pairs nested within 309 tasks and 204 recorded components. Its both-full-input-sufficient stratum contains 12,067 pairs within 308 tasks and 203 components. At low capacities this stratum does not require each input's own truncated prefix to be sufficient. The stratum is selected by auditor references, after the response bank exists; its means cannot be treated as qualification yield or as an unbiased sample of useful skills. Components describe recorded structural grouping, not guaranteed semantic independence.

For pair-pooled weighting, each pair receives equal mass. For equal-task weighting, first average pair changes within each task, then average tasks. For equal-component weighting, additionally average task means within each component, then average components. These summaries differ whenever pair counts or task counts differ. In the sufficient-input stratum, cap-one gains 565 and losses 1,409 give -844/12067=-0.0699428193 approximately. The predeclared equal-component result is +3363459013/266277211200=+0.0126314190 approximately; the equal-task result is approximately +0.01013. The sign change is descriptive sensitivity to unit weighting, not a numerical inconsistency.

A new teaching example makes the mechanism transparent without claiming to reproduce the bank: component X has one task with 100 pairs and pair contrast -0.20; component Y has one task with 10 pairs and contrast +0.30. Pair pooling gives (-20+3)/110=-17/110, approximately -0.15455. Equal-component weighting gives (-0.20+0.30)/2=+0.05. Different probability measures answer different questions even on identical rows. The actual bank's exact fractions above are saved results; these two invented components are only an arithmetic explanation.

Computational size must remain separate from independent sample size. Seven caps create 159,558 pair-capacity contracts, two operators create 319,116 outputs, and three endpoints create 957,348

score cells. None creates an additional task or independent history. A future inferential plan would declare whether histories and source families are fixed or sampled and account for both dependencies. The current exact enumerations support finite descriptive claims without a population confidence interval or power guarantee.

$$d_t = \frac{1}{n_t} \sum_{i=1}^{n_t} (S(P_{ti}) - S(B_{ti})) \tag{11}$$

$$d_{\text{component}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|T_c|} \sum_{t \in T_c} d_t \tag{12}$$

$$d_{\text{pooled}} = \frac{\sum_t n_t d_t}{\sum_t n_t} \tag{13}$$

| Unit | Count | Meaning |
|---|---|---|
| Evidence-bearing task | 309 | Fixed exposed reference task |
| Recorded component | 204 | Structural aggregation unit |
| Unique stored input | 2494 | Authored ID response, not a model trial |
| Ordered pair | 22794 | Includes reversed and identical-input views |
| Sufficient-input stratum | 12067 / 308 / 203 | Pairs / tasks / components; not Q qualification |
| Independent histories / qualified pairs | 0 / 0 | Behavioral effect remains null |

Table 3: Do not pool these denominators or relabel computational cells as new independent data.

## 9 S9. Complete capacity results and endpoint tradeoffs

The following table restores pass counts and ties that the compact main table omits for space. Every row retains the same 22,794 pairs. At cap one the pooled full-containment change is (565-2700)/22794; at cap two it is (95-25)/22794; at cap three it is -2/22794. These pair-pooled fractions are not the equal-component primary summaries. For example, cap-three equal-component difference is -1/13056, demonstrating why the displayed transition count is not itself the component average.

The full-containment boundary and the first-three plateau must be read together. For k>=3, first-three evaluates the same three leading IDs at every generated cap, so its zero-gain/two-loss pattern persists through six. Full containment sees newly appended IDs and has no transitions at caps four through six on this bank, although both methods' absolute pass counts rise. Identical scores between methods do not mean all tasks pass: even cap six leaves 2,255 union-unreachable pairs. A complete union cannot create a missing reference ID.

Exact-annotated changes follow a different pattern: at cap one they coincide with sufficient changes for this bank; at cap two promotion has 66 exact gains and no exact losses; from cap three onward all exact comparisons tie. The contract, not the theorem, determines this pattern. The S4 rank proof is deliberately not cited as an exact-match guarantee. Original annotations may omit semantically valid evidence, so the result is phrased as exact reference agreement rather than truth recovery.

The inherited agreement-only audit further shows why an apparently favorable score can be misleading. Against left-union, intersection-only has 13,855 sufficient losses while also having 2,328 exact gains and 22 exact losses. Under equal-component weighting it emits sufficient/exact scores 0.290889/0.150539, compared with a pairwise output-length-matched uniform control at 0.356638/0.221823. It outputs fewer IDs, and the matched random control makes the resulting score difference interpretable. None of these proportions is a success rate for newly executed skills.

| Cap | Gain | Loss | Tie | RR pass | Promotion pass |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 22794 | 0 | 0 |
| 1 | 565 | 2700 | 19529 | 10038 | 7903 |
| 2 | 95 | 25 | 22674 | 18674 | 18744 |
| 3 | 0 | 2 | 22792 | 20306 | 20304 |
| 4 | 0 | 0 | 22794 | 20506 | 20506 |
| 5 | 0 | 0 | 22794 | 20536 | 20536 |
| 6 | 0 | 0 | 22794 | 20539 | 20539 |

Table 4: Full-containment transition and absolute pass counts. Counts are inherited unchanged from the saved v19 results.

## 10 S10. Strong baselines and restricted selection rights

Simple controls are strong because they isolate what the proposed signal contributes. Round-robin preserves interleaving without reading references. Stable promotion changes class priority while

matching output count. Intersection-only tests abstaining from noncommon material, but must be compared with an exactly matched-length random subset. Left-priority and reversal controls expose input-order dependence. A negative control may outperform a proposed rule without thereby becoming an independently validated deployment policy. All selection rights and candidate membership remain part of the experiment.

The inherited component-fold selector chooses among exactly four declared rules: left-union, round-robin, agreement-first, and agreement-only. Every sufficient-objective fold selects round-robin; every exact-objective fold selects agreement-only. Thus cross-fitting adds no score through variable selection. Its four-rule per-pair gold oracle reaches 0.840200 sufficient and 0.153696 exact, but the excluded disagreement-first control reaches 0.842416 sufficient. This is not a contradiction: a maximum over one candidate set does not bound a method outside that set. Retrospective admission of the negative control would be a new selection decision.

Component-held-out choice makes local access rights clear but cannot undo prior exposure of the annotations, tasks, and diagnostic design. Cawley and Talbot [21] is cited only within its recorded metadata/abstract scope; we transfer no unread full-text theorem. Self-consistency [20] is a separate final-answer aggregation mechanism with sampled reasoning paths, not a native implementation of our sentence-ID intersection. ERASER [4] supplies a rationale-evaluation distinction, not proof that an annotation-contained output is faithful to a model. SkillsBench [19] is the versioned release artifact, not a paper performance baseline run here.

Future native or inspired comparisons need their own explicit contract. Conservative replay must be allowed to reject, narrow, split, or fall back; an always-merge straw controller would suppress its strength. Typed migration needs caller/resource updates and behavioral acceptance beyond schema legality. Runtime routing/assembly must be charged for its decisions and any LLM calls. A fixed-pool adapter and the full native system answer different questions: identical candidate bytes may be appropriate for the first but remove a native system's candidate-generation mechanism from the second. No imported score is placed beside the local finite counts.

| Comparison | Strength that must be retained | Current status / fairness requirement |
|---|---|---|
| RR, stable promotion, intersection and uniform controls | Simple count/order/length mechanisms | Local deterministic diagnostics; no agent inference |
| SkillCommit-inspired replay [14] | Reject, narrow, split, fallback | Not run; equal source feedback and fully charged replay |
| SkillSmith-inspired migration [15] | Typed edits plus caller/resource repair | Not run; equal rights and behavioral checks |
| SkillCAT-inspired assembly [16] | Runtime selection and assembly | Not run; charge router/assembler and do not relabel concatenation |
| SkillEvoReg-style regression [17] | Candidate-conditioned stress validation | Not run; stress failures are not unselected prevalence |
| Signal or Noise / SkillsBench [18,19] | Instruction/resources controls; native workflow packages | Not run; tokenizer, assets, environment and task rights must be frozen |

Table 5: Comparison mechanisms and their current execution status. 'Inspired' does not mean native reproduction.

## 11 S11. Frozen execution and provenance ledger

The canonical unchanged analysis directory is outputs/research_advancement_v19_20261019/skillmerge_analysis_01. Its input_lock.json records UTC lock time 2026-10-19T07:14:31.340185+00:00, eleven original paths, frozen input copies, byte sizes, and full SHA-256 digests. A later result timestamp follows that lock. This is pre-execution freezing of the follow-up, not a claim that the author had never inspected the earlier exposed bank. The earlier primary and post-primary priority plans are preserved as distinct records.

The result binding is results.json SHA-256 e495f64342f8d22c47d39b94f61137e35c9f85f3d0fefa82977aa9df349393af. The frozen plan SHA-256 is 8abf1eec6cd63862acf858c9c5da0724eface01265eab6432c9c7e00fcc88624; analysis code SHA-256 is eef8bd5769cd9f024cbaea5d62f1cce116639488a6d70a9f2dafeede1f3dd513. The input-lock file itself is bound by 5c15bfafeb079f718927e5e503903b65dbf5438457d34341580fc84d042ecdf1. These are artifact identities, not checksums of scientific validity. Abbreviated input prefixes in the table are navigation aids only; integrity comparison uses the complete digests in the lock.

The analysis ledger reports one analysis run, two verification attempts, one failure and one success. The outputs comprise 63 aggregate rows, 3,387 sufficient transition events, and the nested pair/capacity/endpoint counts in S8. Existing inputs remain unchanged. New independent units, model

predictions, actual A/B histories, Q-qualified pairs, native benchmark runs, and model API calls are zero. Behavioral effect and measured token cost are null; a zero count of model calls does not measure the human/Codex writing effort, which is not instrumented.

The following commands describe reproduction into new directories; they were not executed for this compact editorial revision. Analyzer: python -X utf8 -B projects/research_advancement_v19_20261019/skillmerge/analyze.py –out NEW_EMPTY_ANALYSIS_DIRECTORY. Checker: python -X utf8 -B projects/research_advancement_v19_20261019/skillmerge/verify.py –run NEW_EMPTY_ANALYSIS_DIRECTORY –out NEW_EMPTY_VERIFICATION_DIRECTORY. Select actual fresh paths before use; do not overwrite the canonical run or the retained failed verifier. Python standard-library execution is different from a native-model evaluation and does not authorize an external API request.

| Slot | Frozen snapshot | Role | SHA-256 prefix |
|---|---|---|---|
| 0 | 0_0_task_measurements.json | Task/reference measurements | 5c064d56419342da |
| 1 | 1_pair_rows.jsonl | Saved ordered outputs | a929700b78786ae2 |
| 2 | 2_results.json | Primary agreement aggregates | 83550378a0d670ae |
| 3 | 3_results.json | Controlled-priority aggregates | defe151ec17fdfab |
| 4 | 4_witnesses.json | Saved priority witnesses | 1724895a024e806c |
| 5 | 5_PLAN.json | Primary plan | 7716afd2945e5191 |
| 6 | 6_PRIORITY_PLAN.json | Post-primary priority plan | 86e020fcce944b27 |
| 7 | 7_analyze.py | Primary construction code | 79c10dccd758ddca |
| 8 | 8_priority_audit.py | Controlled-priority code | 9fb90db9a42d639a |
| 9 | 9_ISSUES.json | Open scientific boundaries | 7481133c58f1f064 |
| 10 | 10_CLAIMS.json | Prior claim inventory | 62587bbfcc51f278 |

Table 6: The eleven frozen consumed inputs. Full original paths, byte counts and digests are in the canonical input_lock.json; digest prefixes here are explicitly abbreviated.

## 12 S12. Failed verification and the corrected claim

The initial frozen verifier tested a statement stronger than the registered theorem: for all individually sufficient inputs, the promoted output must pass at cap at least four. S6's disjoint-input example refutes that stronger statement because promotion leaves both failing prefixes unchanged. The intended theorem only excludes a baseline-success-to-promotion-failure transition. A failed check can therefore reveal a verifier specification error rather than a reason to change or hide the underlying analysis. Here the original failure is part of the public evidence trail.

The frozen initial verifier digest is 27b8657b52b0d675d150c16e66bbd765b6dedfd2cfda05662a9a1420bff7f391. That source snapshot remains unchanged. The corrected executed checker is separately frozen and bound by 3f55cfb7520aa4db75765541fe21afee07f691f8d3a15866f2b4319c6d5175e7. Its verifier_lock.json records the narrow correction: check the registered no-loss implication instead of universal success. No plan, source input, analysis output, sufficient-input stratum, or endpoint was changed to make a scientific result pass.

The successful verification_02/verification.json reports eleven input-hash checks, 22,794 pair certificates, 159,558 pair-capacity contracts, 957,348 endpoint cells, 3,387 transition events, 63 aggregate rows, 609 risk-stratum rows, and 857,997 authored reference-capacity cells. Separately written prefix-set logic and bitmask enumeration avoid importing the analyzer's constructor functions. This provides error-detection diversity within one author's work; it is neither an independent laboratory replication nor an external expert endorsement.

The finite certificate includes zero-cap, empty-common, empty-input, beyond-union-cap, and absent-reference cases. Under nonempty references, an empty input never enters the both-sufficient class. Equal or infinite completion ranks tie; source families with no fully reachable alternative never become successes by reordering. The corrected no-loss implication is consistent with unchanged failures in these cases. Historical analysis and verification commands are retained in VERIFICATION_HISTORY.json; the compact rewrite does not rewrite a failed attempt as a successful original execution.

| Attempt | Predicate | Outcome | Changed scientific data? |
|---|---|---|---|
| 1 | Unconditional promoted success at cap>=4 | Failed on disjoint sufficient inputs | No |

| Attempt | Predicate | Outcome | Changed scientific data? |
|---|---|---|---|
| 2 | Baseline success implies promoted success | Passed registered finite checks | No; checker assertion corrected only |

Table 7: Correction scope: a failed specification check is retained, with an explicitly frozen replacement checker.

## 13 S13. Prospective independent-history protocol

This section is an unexecuted feasibility design inherited from the research program, not additional evidence. An event fixes a common ancestor, source family, resources, tool rights, and independently authored branch changes A/B. Authors may inspect development D but neither each other's patch nor auditor-only Q/S labels. Freeze proposal caps, duplicate detection, and proposal order before qualification. Qualification is an observed isolated-branch gain against the same ancestor on Q, not a role assigned by a critic and not the annotation condition of the current stored input stratum.

The bounded feasibility plan proposes three independently sourced families and four attempted A/B histories per family: twelve attempts, not twelve guaranteed eligible pairs or a powered confirmatory sample. Each branch has at most two proposals. On twenty source-disjoint Q tasks and one predeclared seed, require each branch to exceed the ancestor by at least two successful tasks. Freeze all proposals before Q, return no Q feedback to authors, and select the first passing proposal in fixed order. At most five configurations, ancestor plus four proposals, imply at most 100 task-arm evaluations per history; these are not necessarily 100 API requests, and generation/tool costs remain additional.

An eligible pair would receive fresh reset S0/A/B/AB evaluations on twenty previously unexposed S tasks with two predeclared seeds: at most $4\times20\times2=160$ final task-arm-seed evaluations per eligible pair. The target compares joint availability against an equal mixture of isolated branches, not a best branch picked after viewing S. Source-family/history weighting must be fixed before S. If no history qualifies, retain yield zero and the behavioral effect null; weakening the Q gate after observing AB would define a different target.

An interaction contrast is not the same target. In a teaching cell (S0,A,B,AB)=(0,1,1,1), the interaction AB-A-B+S0 is -1 but the joint-minus-mean-isolated contrast is zero. In (1,1,1,0), that primary contrast is -1. These arithmetic cells are not assigned observations. Trace availability, actual routing, actual reading, and assembled bytes separately, because installing a skill is not the same intervention as forced activation or exposing its contents to the final model.

Before any real model call, separately specify the actual model identity, allowed transmitted data, request/token/cost limits, tokenizer, reset policy, timeouts, and deployment endpoint. Rejected proposals, repairs, loading/cache, routing, assembly, retries, and final runs must be charged. Conservative baselines may fall back or defer, but completion, deployment coverage, deferral and total cost must be reported together. A policy that refuses every difficult task is not unconditionally successful. Candidate-conditioned stress cases, mixed feedback rights, or a revised candidate class require their own labels and comparisons.

Admit answer expressibility, references, endpoint, source rights, reset policy and full cost accounting.
Freeze ancestor/resources, source-disjoint D/Q/S, model identity, proposal cap and candidate class.
Independently author capped A/B proposals using D only; retain every failure and duplicate attempt.
Apply the frozen Q gain gate with no author feedback; choose first passing proposals by fixed order.
If eligible count is zero, report yield zero and behavioral effect null.
Freeze selected bytes and canonical AB order; run fresh assigned S0/A/B/AB resets and predeclared controls.
Keep missingness, deferral, nonactivation and resource traces; charge all policy and generation work.
Estimate only the declared contrast and weighting; perform attribution through matched controls.

Algorithm 2: Prospective gates only. This editorial revision and the saved deterministic audit execute none of these model/history steps.

## 14 S14. Admission decisions and falsification ledger

The current audit supports an engineering admission decision, not deployment of the highest finite score. Before treating a negative result as semantic interference, check that the response can express a valid answer, that all original alternatives are preserved, that the grader applies the declared ranking/cap, and that the comparison changes only its intended mechanism. If a same-set reordering changes a score, that is evidence about the ranked contract; it does not by itself establish loss of full evidence. If a merge actually omits a required ID under a cap, omission is a narrower observed mechanism than general semantic conflict.

Falsification must be attached to a claim. The explanation that joint use lost annotated evidence is weakened when complete output sets retain it and only first-three fails. The explanation that

common-item priority is beneficial in general is refuted by the saved sufficient-input witness; the stronger universal no-loss claim for unrestricted changed sets is refuted by S5's singleton construction. The conditional cap-four theorem survives those examples because it states narrower assumptions. A prospective natural-skill effect would require fresh histories and could be weakened if its negative contrast disappears under matched ordering, actual tokenizer length, information rights, unchanged resources, and the frozen endpoint.

The future decision ledger must retain assigned availability cells, invalid outputs, agent-caused timeouts, nonactivation, deliberate deferral, and unresolved external outages. External outages need an explicit missingness policy and bounded summaries rather than silent deletion or automatic success. Repeated seeds share histories; repeated tasks may share source families. Qualification yield, paired variability, and feasible cost from an independently sourced pilot should inform later precision planning, without turning the current zero-history ledger into a power calculation.

The final compact conclusion is intentionally narrower than a broad skill-composition claim: under a specified finite annotation contract, stable promotion can displace completion, a rank threshold determines its capacity window, and declared weighting can reverse an aggregate sign. The complete historical manuscript remains in inherited.paper to preserve prior diagnoses, sensitivity envelopes, comparison discussions, and source scopes. Historical version progression is an audit trail in the supplement, not a stack of separate claims of new behavioral evidence.

| Claim or decision | Current warranted statement | Additional requirement |
|---|---|---|
| Capacity mechanism | Exact finite transitions and conditional no-loss proof | New assumptions require new proof/ checks |
| Reference score recovery | Endpoint-specific operational recovery | Independent semantic adjudication for truth claims |
| Agreement advantage | Length/order controls can reverse interpretation | Fresh candidate class and equal information/cost |
| Behavioral composition harm | Null effect; zero independent qualified histories | Independent authoring, observed Q gate, reserved S |
| Deploy or promote a skill | Not authorized by these deterministic scores | Separate behavior/cost validation and deployment decision |

Table 8: What the evidence permits now, and the additional observation needed for a stronger decision.

## 15 S15. The conditional reference class and label-blind switch

This section distinguishes four objects often collapsed into one word, useful: a reference alternative contained by a fixed output; containment by the complete input before truncation; independently measured skill improvement on Q; and deployment correctness. Only the first two enter the new mathematics. A and B are ordered, distinct-ID input lists. Their complete union order V alternates A then B by position and removes repeated occurrences. P is a stable partition of V into common IDs followed by noncommon IDs. No reference, task label, prior score, or judge feedback enters either order.

Fix A, B and a nonnegative integer cap k before choosing a reference family. X and Y are the ID sets of the equal-size capped outputs V_k and P_k. Consider every finite nonempty family of nonempty reference alternatives having at least one alternative contained in full A and at least one contained in full B. The two witnessing alternatives may coincide or overlap; they need not exhaust the family or the input. Extra alternatives may use IDs outside the union. The forward guarantee covers all such families. We do not assume the annotations enumerate every semantically correct answer.

The guard accepts if X=Y, or all of A is retained in Y, or all of B is retained in Y. It compares IDs and sets only. The guarded output is exactly P_k on acceptance and exactly V_k on rejection. This is a cap-dependent switch between two complete candidate outputs, not an LLM router, skill assembler, label-aware oracle or a task-specific learned selector. An observed both-sufficient flag is computed later to describe which rows satisfy the theorem; it is not an input to the guard.

Two quantifier orders must not be exchanged. The theorem fixes A/B/k and asks whether promotion is safe for every admissible reference family. It does not inspect the actual family and find the largest set of acceptable promotions. The latter task could use an actual reference-aware certificate for a retrospective audit, but exposing that certificate to a deployed constructor changes its information rights. No such oracle policy is scored as a reference-blind method here.

$$X = \mathrm{set}(V_k), \quad Y = \mathrm{set}(P_k) \tag{14}$$

$$G(A, B, k) = \mathbb{1}[X = Y \vee \mathrm{set}(A) \subseteq Y \vee \mathrm{set}(B) \subseteq Y] \tag{15}$$

$$H_k = \begin{cases} P_k G = 1 \\ V_k G = 0 \end{cases} \tag{16}$$

Build the complete deduplicated alternating order V using A/B only.
Let C=set(A) intersect set(B); P is V restricted to C followed by V restricted to its complement.
Take equally long prefixes V_k and P_k, with length min(k,len(V)).
Accept if their ID sets coincide, or P_k contains every ID of A, or P_k contains every ID of B.
Return P_k on acceptance; otherwise return V_k and record a fallback.
Only after construction, read the original reference family to score endpoints and record descriptive strata.

Algorithm 3: The full selection rule. Freeze the outputs before any reference-dependent operation. The guard is not supplied a sufficiency flag.

## 16 S16. Exact conditional theorem and reverse counterfamily

Theorem. For fixed A/B/k and equal-cardinality X,Y contained in the same union, the guard is true if and only if full-containment success of X implies success of Y for every reference family in the conditional class of S15. The algebraic theorem is not restricted to stable promotion; the concrete operators provide the particular X,Y tested. Nonempty alternatives exclude automatic success from an empty rationale. This statement promises no-loss, not successful output on every task.

Forward, case one: X=Y. Every alternative is either contained in both sets or neither, so scores coincide. Forward, case two: A is wholly retained in Y. By membership in the conditional class, some alternative R_A is contained in A. Transitivity puts R_A inside Y, so Y passes regardless of whether X passes. Case three for B is identical. This proof allows unequal input lengths, overlapping alternatives, more than two alternatives, incomplete reference inventories, and extra alternatives outside the union. None of those variations changes the implication.

Reverse: suppose all guard clauses are false. Because X and Y have the same finite cardinality and differ, there is an x in X outside Y. Since neither full input is inside Y, choose a_out from A outside Y and b_out from B outside Y. If x belongs to A, use {x} as the A-witnessing alternative; otherwise use {a_out}. If x belongs to B, use {x} as the B-witnessing alternative; otherwise use {b_out}. Remove duplicate singleton alternatives if they coincide. At most two nonempty alternatives remain. Each complete input contains one, at least one equals {x}, and none is inside Y. Therefore X succeeds and Y fails on this admissible family. This explicitly refutes universal conditional safety whenever the guard rejects.

The reverse construction proves restricted maximality. A binary switch that sometimes emits P_k and otherwise emits V_k cannot safely accept any rejected shape while preserving this universal conditional promise: that shape has the constructed counterfamily. Maximality is relative to both the two-candidate policy class and the quantified reference class. It is not Pareto-optimality under the actual annotations, semantic correctness, expected utility, token cost, or arbitrary new output construction. Actual reference knowledge could safely accept more shapes, and the saved bank contains two gains that this conservative guard declines.

Equal cardinality is used only to turn X!=Y into an ID removed from X. With unequal sizes, replace the equality clause by X subset Y. If no ID is removed, monotonicity of full containment gives the forward case; if an ID is removed and neither complete input is retained, the same singleton construction works. We do not implement or score an unequal-budget policy. The equality-based guard must not be copied to a different-length output comparison without revisiting that premise.

At k=0, X=Y is empty and the guard accepts. With nonempty alternatives both scores fail, hence no-loss holds without success. If an input is empty, no nonempty reference family can make it sufficient; the conditional class is vacuous. For the present stable constructor, an empty input also leaves P=V and satisfies the guard, so no empty eligible population is manufactured. An empty common set leaves P=V at every cap. These endpoints are covered explicitly by the separate verifier.

$$G = 1 \Leftrightarrow \left(\forall \mathcal{R} \in \mathcal{F}_{A,B} : S(X) = 1 \Rightarrow S(Y) = 1\right) \tag{17}$$

$$x \in X \setminus Y, \quad a_{\text{out}} \in \text{set}(A) \setminus Y, \quad b_{\text{out}} \in \text{set}(B) \setminus Y \tag{18}$$

## 17 S17. Sharp general-length boundary and authored counterexamples

Corollary. If both input lengths are at most d>=2 and the stable partition is formed from the complete alternating union before capping, every cap k>=2d-2 satisfies the structural guard. Conditional no-loss follows from S16. This generalizes length three/cap four without silently removing its assumptions. It is a uniform worst-case bound: particular shapes can be accepted at a much smaller cap, which is the reason to compute the actual guard.

Proof by common-set size. With no common ID the partition leaves V unchanged, so X=Y. With at least two common IDs, union size is at most 2d-2 and the cap retains the whole union. With exactly one common ID, union size is at most 2d-1, hence at most one ID is omitted from P_k. That one common ID appears first; any omitted ID is exclusive to one input. The other complete input is retained. In each case a guard clause is true. This uses full-union construction and stable relative order when identifying the concrete baseline and promoted outputs, and it does not license a grader that ignores later IDs.

Sharpness for every integer d>=2 follows from two lists of length d. Put d-1 private A IDs and d-1 different private B IDs first, then the same common ID last in each. Their alternating union lists the private IDs in A/B order and ends with the common one. At cap 2d-3, RR includes the last private A ID but not the last private B ID. Promotion spends one slot on the common ID and includes neither last private ID. Use those two private singleton IDs as alternatives. Each original input is sufficient, RR completes one alternative, promotion completes none, and the guard rejects. At cap 2d-2, the other complete input is retained and the guard accepts.

The seven d=2..8 saved examples are authored illustrations of that symbolic construction, not seven sampled tasks or external evidence. The proof establishes the all-d statement. For d=0 and d=1, check nonnegative caps directly: empty or singleton inputs cannot create a changed stable order causing a conditional loss. We do not plug d=0 into an expression that yields a negative output capacity.

The result remains no-loss rather than universal success. As a teaching example, take disjoint A=(0,1,2), B=(3,4,5), with alternatives {0,1,2} and {3,4,5}. At cap four both operators select (0,3,1,4), completing neither alternative. Both complete inputs were sufficient. This same logical distinction caused the preserved v19 verifier failure in S12; the current verifier asserts implications rather than success.

$$|A \cup B| = |A| + |B| - |A \cap B| \tag{19}$$

$$|A| \leq d \wedge |B| \leq d \wedge d \geq 2 \wedge k \geq 2d - 2 \Rightarrow G = 1 \tag{20}$$

| Maximum input length d | Guaranteed no-loss from cap | Constructed loss at cap | Evidence type |
|---|---|---|---|
| 2 | 2 | 1 | General proof; authored check |
| 3 | 4 | 3 | General proof; authored check |
| 4 | 6 | 5 | General proof; authored check |
| 5 | 8 | 7 | General proof; authored check |
| 6 | 10 | 9 | General proof; authored check |
| 7 | 12 | 11 | General proof; authored check |
| 8 | 14 | 13 | General proof; authored check |

Table 9: The sharp worst-case boundary. These symbolic length cases and authored checks are not new benchmark data.

## 18 S18. Guarded capacity results and fallback denominators

All rows below reuse the same exposed ordered-pair bank. Full-containment transitions are paired to round-robin; RR therefore has zero gains/losses against itself. The table keeps both the all-pair cohort and the fixed both-input annotation-sufficient stratum. The latter is a reference-derived descriptive subset, never a policy input or observed Q qualification. Counts of accepted guard shapes and fallbacks sum to the row denominator. Flags can overlap, so separate reason partitions in aggregates.json assign equality first, then full A, then full B, then fallback.

At cap one in the sufficient stratum, promotion has 565 gains and 1,409 losses. The guard retains 563 gains and no losses after 1,502 fallbacks. Of those fallbacks, 1,409 prevent an actual loss, two discard an actual gain, and the remaining 91 tie on full containment. Thus fallback is not synonymous with repaired failure. At cap two, 29 fallbacks prevent 25 losses and retain four ties; all 95 gains survive. At cap three, both fallbacks prevent the two losses, but H has no gains over RR. A repair of promotion is not automatically an improvement over the baseline.

All-pair cap-one fallback is 2,361. It prevents 2,213 promotion losses, sacrifices two gains and leaves 146 score ties. The other 487 promotion losses are accepted because the conditional premise is not met. This arithmetic is compatible with conditional safety and refutes unconditional safety. The not-

both-sufficient cohort has 10,727 pairs from 292 tasks and 193 components; it is neither a disjoint new dataset nor a second independent experiment.

From cap four, H always accepts on the length-at-most-three bank and full transitions all tie. It still changes an ordered output in ways relevant to a prefix-reading evaluator. Accepted guard is not the same as unchanged sequence, and identical selected sets do not imply identical first-three prefixes. All raw method outputs and endpoint scores remain in seven separate pair files. The plot uses the sufficient stratum only; all-pair failure is shown numerically and in S19 rather than silently projected away.

| Cohort | Cap | RR passes | P gain/loss | H gain/loss | H passes | Fallback |
|---|---|---|---|---|---|---|
| All | 0 | 0 | 0 / 0 | 0 / 0 | 0 | 0 |
| Both sufficient | 0 | 0 | 0 / 0 | 0 / 0 | 0 | 0 |
| All | 1 | 10038 | 565 / 2700 | 563 / 487 | 10114 | 2361 |
| Both sufficient | 1 | 7452 | 565 / 1409 | 563 / 0 | 8015 | 1502 |
| All | 2 | 18674 | 95 / 25 | 95 / 0 | 18769 | 34 |
| Both sufficient | 2 | 11666 | 95 / 25 | 95 / 0 | 11761 | 29 |
| All | 3 | 20306 | 0 / 2 | 0 / 0 | 20306 | 2 |
| Both sufficient | 3 | 12060 | 0 / 2 | 0 / 0 | 12060 | 2 |
| All | 4 | 20506 | 0 / 0 | 0 / 0 | 20506 | 0 |
| Both sufficient | 4 | 12067 | 0 / 0 | 0 / 0 | 12067 | 0 |
| All | 5 | 20536 | 0 / 0 | 0 / 0 | 20536 | 0 |
| Both sufficient | 5 | 12067 | 0 / 0 | 0 / 0 | 12067 | 0 |
| All | 6 | 20539 | 0 / 0 | 0 / 0 | 20539 | 0 |
| Both sufficient | 6 | 12067 | 0 / 0 | 0 / 0 | 12067 | 0 |

Table 10: Full-containment transitions versus RR. All rows have 22,794 pairs or 12,067 sufficient-input pairs. Counts are computed reuses, not independent integration histories.

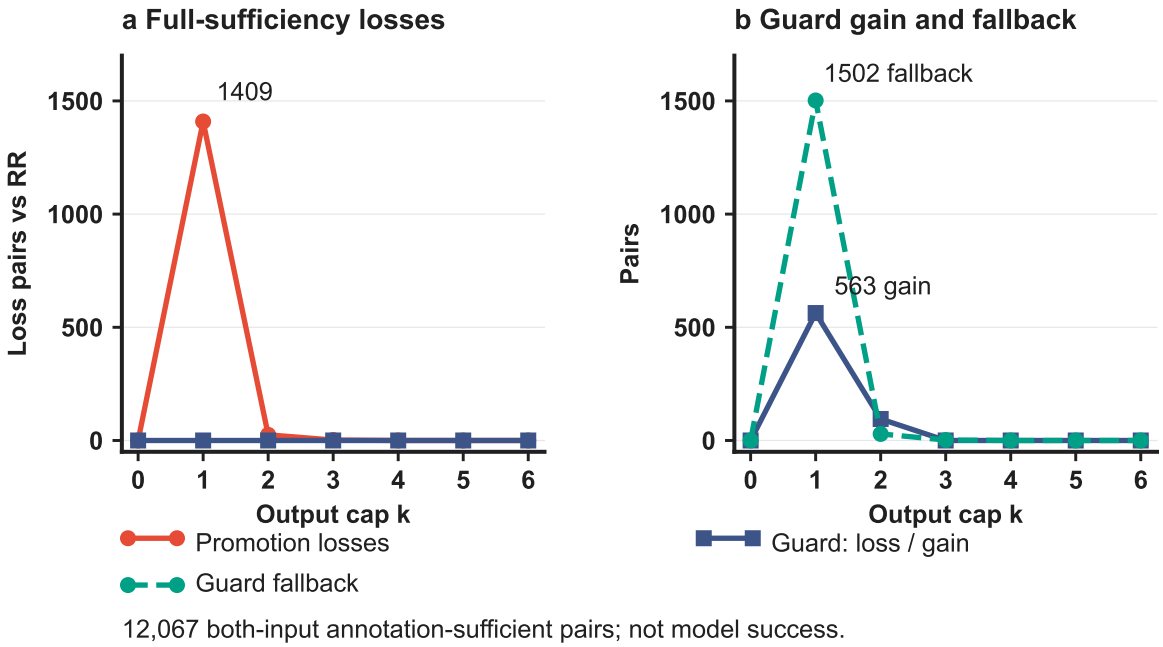



Figure 1: New guarded-promotion audit on the fixed both-input annotation-sufficient stratum: full-containment gain/loss versus RR and fallback counts by output cap. Every cap reuses 12,067 pairs, 308 tasks and 203 components. The guard loses no full-containment score in this conditional stratum but discards two cap-one gains; at cap three it ties RR. This figure does not include or erase the 487 all-pair cap-one losses, first-three failures, or absent behavioral skill outcomes.

## 19 S19. Saved gains, conservative rejection and unconditional failure

The saved examples in witnesses.json are actual rows of the existing constructed bank. They are not newly generated model outputs. Task scifact_v4_39 has singleton reference alternatives {7},{8},{9},{10},{11}. With A=(0,7), B=(7), cap one, retaining complete B accepts promotion (7), changing RR failure (0) into success. With A=(7,0), B=(0,8), both inputs contain a reference alternative, but

promotion (0) loses RR success (7). Neither full input is retained and sets differ, so the guard falls back and blocks the loss.

The very same policy also preserves an actual loss outside the premise. On scifact_v4_39 with A=(7,0), B=(0), cap one, retention of complete B accepts (0); RR would output (7). Only A is annotation-sufficient. The smallest authored shape of this failure uses A=(0,1), B=(1), reference {0}, as shown in S16-S17. Its union size two, total input length three and cap one are minimal under the finite search ordering stated in PLAN. Minimality is within that declared finite construction, not over all possible agent tasks.

Conservatism is visible on scifact_v4_653. Its singleton alternatives are {0},{1},{5}. With A=(2,0), B=(0,1), cap one, RR selects (2) and fails; promotion selects (0) and passes. Both complete inputs are sufficient, but neither complete input fits the promoted prefix and its set differs from RR. The guard rejects. This is one of the two sacrificed gains. No contradiction follows: a different admissible family using {2} and {1} would make this same shape a promotion loss, so universal safety must reject it.

The last contrast explains why an actual family and a universal class give different answers. We do not alter gold to boost or reduce a measured result: the counterfamily is an authored proof witness, explicitly separate from the stored task references. Its purpose is to show why a reference-blind policy cannot both accept that shape and keep the universal conditional promise. Model-generated text, unannotated semantic validity and skill usefulness need different observables.

| Case | A ; B | Reference family | RR ; P ; H at cap1 | Meaning |
|---|---|---|---|---|
| Saved gain | (0,7) ; (7) | {7},{8},{9},{10},{11} | (0) ; (7) ; (7) | Both sufficient; safe acceptance |
| Saved blocked loss | (7,0) ; (0,8) | {7},{8},{9},{10},{11} | (7) ; (0) ; (7) | Both sufficient; fallback |
| Saved remaining loss | (7,0) ; (0) | {7},{8},{9},{10},{11} | (7) ; (0) ; (0) | B insufficient; outside premise |
| Saved discarded gain | (2,0) ; (0,1) | {0},{1},{5} | (2) ; (0) ; (2) | Both sufficient; conservative loss of gain |
| Authored counterfamily | (2,0) ; (0,1) | {2},{1} | (2) ; (0) ; (2) | Proof only; not original task gold |

Table 11: Saved rows and one explicit proof-only counterfamily. Parentheses denote ordered outputs; braces denote alternatives. Original task references are never replaced in scoring.

## 20 S20. Endpoints, cap-dependent switching and weighting

The universal theorem concerns monotone full containment. First-three observes an ordered subprefix, not the equal-cardinality sets compared by G, while exact-annotated can reject a sufficient superset. Each endpoint is recomputed separately for every method. On this bank H has 563/0 first-three gain/loss in the sufficient stratum at cap one, 95/0 at two, 0/0 at three, and 0/2 at four through six. The cap-four losses are not a theorem failure; the evaluator reads a different object.

The first successful capacity for each fixed complete order V or P is its minimum rationale-completion rank, as proved in S4. H is a switch whose choice can change with k. Therefore its outputs are not automatically prefixes of one fixed global order, and the earlier single-window proof must not be imported without checking nestedness. Our new raw rows allow direct cap-by-cap validation and make no universal single-window claim for H.

Weighting describes a different unit of averaging, not a correction that turns constructed multiplicities into natural prevalence. The exact aggregate means stored in results.json are absolute pass rates; a contrast requires subtracting the RR mean with the same weighting. In the sufficient stratum, H pooled cap-one contrast is 563/12067; its equal-component contrast is 124638679619/1757429593920. At cap two they are 95/12067 and 106184173/6051754800 respectively. Their positive signs follow from zero paired full losses here, not from independent evidence.

For unprotected promotion the preserved cap-one sufficient-stratum contrast still reverses sign: pooled -844/12067, equal-component +3363459013/266277211200. All-pair guarded cap-one pooled contrast is only 76/22794, despite 563 gains, because 487 losses remain. Reporting the gains alone would hide the conclusion-changing denominator and countervailing transitions. The table reports H endpoints with separate cohort denominators; native benchmark percentages are not mixed into it.

| Cohort | Cap | Full gain/loss | First-three gain/loss | Exact gain/loss |
|---|---|---|---|---|
| All | 0 | 0 / 0 | 0 / 0 | 0 / 0 |
| Both sufficient | 0 | 0 / 0 | 0 / 0 | 0 / 0 |
| All | 1 | 563 / 487 | 563 / 487 | 563 / 487 |

| Cohort | Cap | Full gain/loss | First-three gain/loss | Exact gain/loss |
|---|---|---|---|---|
| Both sufficient | 1 | 563 / 0 | 563 / 0 | 563 / 0 |
| All | 2 | 95 / 0 | 95 / 0 | 66 / 0 |
| Both sufficient | 2 | 95 / 0 | 95 / 0 | 66 / 0 |
| All | 3 | 0 / 0 | 0 / 0 | 0 / 0 |
| Both sufficient | 3 | 0 / 0 | 0 / 0 | 0 / 0 |
| All | 4 | 0 / 0 | 0 / 2 | 0 / 0 |
| Both sufficient | 4 | 0 / 0 | 0 / 2 | 0 / 0 |
| All | 5 | 0 / 0 | 0 / 2 | 0 / 0 |
| Both sufficient | 5 | 0 / 0 | 0 / 2 | 0 / 0 |
| All | 6 | 0 / 0 | 0 / 2 | 0 / 0 |
| Both sufficient | 6 | 0 / 0 | 0 / 2 | 0 / 0 |

Table 12: Guarded transitions versus RR by endpoint. These are finite descriptive counts, not confidence intervals or independently sampled outcomes.

## 21 S21. New execution lock and separate author verification

The v21 canonical run is runs/research_advancement_v21_20261019_01/skillmerge/analysis_01. PLAN.json, analyze.py, verify.py and six consumed sources are copied and hash-fixed before computation. Discovery follows the preserved v19 CURRENT_RESEARCH pointer; the new calculation uses that run's copied task measurements and pair rows rather than fetching a changing dataset. The earlier v19/v20 project and run bytes are not mutated. v21 output existence is not permission to replay into it: every rerun needs a fresh output directory.

The six frozen consumed sources are the v19 discovery pointer, input lock, task measurements, pair rows, analyzer code, and results. input_lock.json records complete workspace-relative paths, bytes, hashes and freezing time. The output includes pairs_cap_0.jsonl through pairs_cap_6.jsonl, aggregates.json, finite_certificates.json, witnesses.json, results.json and receipt.json. Every pair file is below the text-resource size limit. NEW_RESEARCH.json binds the success result, plan, lock, proof, verifier and literature review for publication. Hash equality certifies those bytes, not semantic correctness or independent qualification.

The separately written checker does not import analyzer functions. It independently schedules first occurrences, applies a stable class sort, computes bitmask endpoints, and reconstructs all 159,558 pair-cap rows and all 21 cohort-cap aggregates, including exact Fraction means and overlapping guard flags. It then checks finite conditional safety and explicit reverse witnesses. Verification is authored by the same agent, so it is a cross-implementation consistency check rather than external replication. One v21 analysis and one verification attempt complete successfully; earlier failed v19 checks remain separate and are not relabeled as v21 successes.

Finite construction uses all distinct-ID input permutations up to length three whose union is each enumerated universe, including endpoint cases. There are 3,046 input pairs and 21,322 pair-cap shapes. G accepts 19,210 shapes and rejects 2,112; every rejected shape gets an explicit conditional counterfamily. The 777,333 accepted checks enumerate ordered nonempty subsets of the two inputs as witnessing alternatives. Different ordered subset pairs can represent the same family. Consequently that number is not a count of distinct reference families, tasks, independent samples or skill events.

Expected tamper rejection alters separately copied input and output files under verification_01; it never rewrites original frozen evidence. Both mismatches are detected. The verification JSON has ok=true and binds results SHA-256. Unexpected failure would have required preserving its directory and a separately frozen correction; no such v21 correction was needed. Execution commands below use Python standard library only. The Codex writing/reasoning process is not instrumented as experimental model cost and is not falsely counted as zero writing effort.

| Unit | Count | Interpretation |
|---|---|---|
| Ordered pairs / tasks / components | 22794 / 309 / 204 | Exposed fixed bank |
| Pair-cap decisions | 159558 | Seven repeated capacities |
| Deterministic outputs | 478674 | Three policies per pair-cap |
| Endpoint cells | 1436022 | Three endpoints per output |
| Aggregate rows | 21 | Three descriptive cohorts by seven caps |
| Finite shapes / reverse witnesses | 21322 / 2112 | Authored mathematical checks |
| Ordered subset-pair checks | 777333 | Duplicate family representations possible |
| Model/API / independent histories / Q pairs | 0 / 0 / 0 | Behavioral effect null |
| Native benchmarks / new dependencies | 0 / 0 | Reading-only external resources |

Table 13: Executed v21 accounting. Counts across rows have different units and must not be added. Full paths and hashes are available in NEW_RESEARCH.json and the run locks.

```
python -X utf8 -B projects/research_advancement_v21_20261019/skillmerge/analyze.py –out runs/
research_advancement_v21_20261019_01/skillmerge/analysis_01
python -X utf8 -B projects/research_advancement_v21_20261019/skillmerge/verify.py –run runs/
research_advancement_v21_20261019_01/skillmerge/analysis_01 –out runs/
research_advancement_v21_20261019_01/skillmerge/analysis_01/verification_01
```

Algorithm 4: Executed commands, abbreviated only by choosing python on the configured Python 3.11 executable. Existing output directories are preserved; replay must choose new paths.

## 22 S22. Primary reading bridge and benchmark admission

These are directly inspected primary resources for designing a later behavioral study, not new local empirical outcomes. literature/RESOURCES.json records titles, versions, reviewed sections, source URLs, rights, budgets, access/license scope, six reusable reading fields and short selected-excerpt hashes. The three selected verbatim excerpts total 360 UTF-8 bytes; excerpts and metadata fit the 100,000-byte source limit. No external dataset, container or evaluation dependency was downloaded or executed. The inherited 21 bibliography records remain unchanged.

SkillsBench v1.1 is a fixed native task.md package release, already reference [19], not a new paper or independent result. Its task/environment/oracle/verifier division is useful for deciding what the agent may access and what stays evaluator-only. The official pinned release, not a changing README, supplies the reviewed compatibility instructions. Repository Apache-2.0 text was inspected at the pinned commit; third-party task rights were not exhaustively audited. Re-review strengthens version clarity, not empirical evidence. Source: https://github.com/benchflow-ai/skillsbench/releases/tag/v1.1

FollowBench is relevant for distinguishing constraint types and incrementally added requirements. Its hard, soft and consistent-level measures answer different questions; its all-constraint success is not our any-complete-alternative containment. Selected construction/evaluation sections and the author repository were read. Open-ended judging can expose the constraint evolution path to the judge; future information and cost accounting must keep that privilege separate from the generating policy. The repository Apache-2.0 license was checked, while selected upstream task rights still need review. Sources: https://aclanthology.org/2024.acl-long.257/ and https://arxiv.org/html/2310.20410v3 and https://github.com/YJiangcm/FollowBench

When Instructions Multiply introduces ManyIFEval and StyleMBPP. Compatible simultaneous constraints help distinguish true composition burden from impossible requests; held-out task descriptions and combinations motivate a stronger split design. Selected primary sections and author README were inspected. All-instruction success remains a conjunction, different from completion of any reference alternative. The paper states CC BY4.0 for benchmark data; the inspected repository landing view did not establish a standalone code license. Do not infer universal code permission from that data statement. Sources: https://aclanthology.org/2025.findings-emnlp.896/ and https://arxiv.org/html/2509.21051v1 and https://github.com/kenoharada/Multiple-Instructions-Following

Our proposed priority is a small, rights-checked instruction-composition pilot inspired by the compatible-constraint control, followed by a native skill-environment pilot if independent authoring and Q qualification become feasible. This is an admission preference, not an executed benchmark ranking. Either pilot needs a frozen commit, exact subset, source-disjoint D/Q/S, reset state, scorer and total generation/judging cost before outcomes are visible. Published scores are not inserted into the local guard table.

| Reading | Design value | Difference from present result | Before execution |
|---|---|---|---|
| SkillsBench v1.1 release [19] | Native skill/environment/ verifier boundary | No new A/B qualification from skill/no-skill alone | Pin packages, rights, reset and full costs |
| FollowBench ACL2024 | Constraint type and level; hard/soft split | LLM judging is not exact set containment | Fix judge privileges, path access and budget |
| ManyIFEval / StyleMBPP | Compatible requirements; held-out combinations | All-constraints is not any-alternative success | Verify code/data rights; add independent D/Q/S |

Table 14: How to use the newly reviewed sources without changing their evaluation contract. These are prospective design links, not local reproduction claims.

## 23 S23. Prospective behavioral gates and falsification decisions

The guard proof does not identify a behavioral skill effect because actual skill edits need not preserve a two-input evidence union or satisfy annotation sufficiency. Before moving to agents, admit the task

family only if simultaneous requirements are satisfiable, outputs are expressible, verifier rights are allowed, reset is possible, and a meaningful fresh outcome can be observed. Predeclare what counts as a successful individual skill and what is counted when no eligible pair is found. Zero yield is reported as zero eligible pairs and null composition effect, not as proof of safe integration.

Freeze a baseline environment and independently author A/B on D with bounded candidate counts. Q can select or reject proposals under a predeclared improvement gate but cannot silently become training feedback. Freeze selected bytes and the AB order before reserved S. Require equal effective task information, tool privileges, token limits and total proposal/routing/retry/judge accounting across comparison arms. If a native method uses extra context or an assembler, preserve its defining mechanism and charge the additional resource, rather than substituting concatenation and calling it native reproduction.

Use simple RR-like ordering, concatenation, constrained truncation, conservative replay/rejection, typed repair and native routing as distinct comparison rights. A controller that repairs every observed development failure is still a candidate selected on exposed evidence. Component folds diagnose sensitivity; they do not manufacture a new independent S. Fresh behavioral claims need the planned split and arm executions, with failures, deferrals and missingness retained.

Potential falsifiers are concrete. A conditionally sufficient saved pair with H full loss would refute the implementation or theorem assumptions and stop publication. A natural task in which ID retention leaves semantic execution wrong would refute transfer of the contract to that task, without refuting the set theorem. If AB fails only because its requirements conflict, the study must report compatibility failure rather than an interference mechanism. If apparent harm vanishes under a token-matched length control, the narrower capacity explanation wins. If the qualified pair denominator is zero, the targeted behavioral contrast remains undefined.

| Proposed decision | Required evidence | Retained unfavorable outcome |
|---|---|---|
| Transfer the guard to skills | Explicit output contract and truthful input qualification | Wrong contract or failed qualification blocks transfer |
| Claim behavioral composition harm | Independent A/B; observed Q; fresh S arms | Zero yield, ties, missingness and no gain |
| Prefer a controller | Frozen candidate class; same rights; full cost | Conservative rejection and sacrificed benefit |
| Use a public benchmark | Pinned allowed subset; native scorer; reset | License/access gaps or invalid environment |
| Promote an active SKILL | Separate held-out behavior and resource validation | No activation from set certificates alone |

Table 15: Evidence required for stronger decisions. These tests are not executed by the v21 deterministic calculation.

## 24 S24. Exact v21 reading map and preservation boundary

S1-S14 remain byte-equivalent in content to the v20 supplement and therefore preserve its original reading map and stage-specific statements. The complete v19 scientific record is still rendered after the current supplement. The v20 paper, supplement, Korean guide and metadata are also preserved as exact files under preserved_v20/, with a manifest binding their bytes. Preservation does not turn an older result into a replication or a current claim; this section supplies the current map.

Read the compact main to identify the exact conditional promise and its measured limits. Follow S15-S17 for the definition, both directions of the proof and the sharp length boundary. Use S18-S20 to audit the full denominators, conservative rejection and endpoint mismatch. S21 connects the run to the checker and hashes; S22-S23 explain what the external benchmarks can contribute and what fresh evidence is still missing. The Korean prefix derives simple examples before the full preserved v20/v19 guide.

| Main heading | New supplement heading | New Korean guide heading |
|---|---|---|
| Introduction | S15. The conditional reference class and label-blind switch | v21-1. 이번에 추가한 질문과 읽는 지도 |
| Problem and Method | S15. The conditional reference class and label-blind switch | v21-2. 세 가지 검사만으로 고르는 이유 |
| Problem and Method | S16. Exact conditional theorem and reverse counterfamily | v21-3. 필요충분의 역방향을 손으로 만들기 |
| Problem and Method | S17. Sharp general-length boundary and authored counterexamples | v21-4. 길이가 달라져도 성립하는 경계 |
| Results | S18. Guarded capacity results and fallback denominators | v21-5. 563개 이득과 487개 손실을 함께 읽기 |

| Main heading | New supplement heading | New Korean guide heading |
|---|---|---|
| Results | S19. Saved gains, conservative rejection and unconditional failure | v21-6. 안전 때문에 포기한 실제 이득 두 개 |
| Limitations | S20. Endpoints, cap-dependent switching and weighting | v21-7. 같은 집합이어도 앞 세 개는 다르다 |
| Experimental Setup | S21. New execution lock and separate author verification | v21-8. 실행 근거와 다음에 읽을 논문 |
| Related Work | S22. Primary reading bridge and benchmark admission | v21-8. 실행 근거와 다음에 읽을 논문 |
| Conclusion | S23. Prospective behavioral gates and falsification decisions | v21-9. 책을 덮고 답해 볼 질문 |

Table 16: Current exact headings for the new result. STUDY_MAP.json provides these links structurally, while all previous v20 links are retained.

## 25 Complete preserved v19 scientific record

The following complete record preserves the earlier scientific sections, negative results, equations, tables, algorithms, figures and appendices. Its development-stage statements retain their original scope. The current compact conclusions and detailed derivations precede this archived material; retaining an earlier result does not turn it into an independent replication.

SkillMerge: Agreement, Output Length and the Limits of Compositional Repair: From composition diagnostics to agreement, length controls and candidate-class selection

Neutral working paper v19, 2026-10-19. New finite saved-evidence diagnostics; no new independent data, real model/API calls, native-method trial or physical acquisition. Previous v17 evidence and v18 presentation are retained.

Can agreement between two useful skill outputs safely guide their composition? We distinguish this unexecuted behavioral question from an exposed SciFact-derived diagnostic bank. V17 retains 309 tasks, 204 components and 22,794 ordered pairs, executes five label-blind output constructors and controls their output lengths. Intersection-only raises exact-reference score against a longer left-union rule but loses sufficient coverage and is worse than a pairwise length-matched uniform subset on both endpoints. An explicitly post-primary controlled audit isolates common-item priority against the same round-robin baseline: zero sufficient gains and two losses, including a saved pair whose two inputs both contain valid alternative rationales. Five component folds always select round-robin for sufficient containment and intersection-only for exact matching, giving no score gain from changing the rule across folds. A gold-aware oracle is only an upper bound for the four predeclared selector candidates; an excluded negative control can exceed it and is not retrospectively promoted. Alternate author implementations verify 113,970 deterministic outputs and 25,119 exact uniform probability cells. All data are already exposed annotation-derived constructions. Independent A/B histories and qualified pairs remain zero and the behavioral composition effect remains null. The contribution is an auditable agreement/selection mechanism comparison with explicit information and cost boundaries. The continuation gives exact capacity-window identities and a conditional no-loss boundary, while first-three truncation and unit weighting retain important failures.

- A decision-focused formulation of qualified skill composition: joint availability versus the mean isolated branch, conditional on an independently applied qualification gate, with eligibility yield separated from effect.
- An executed evidence-contract audit preserving source connectivity, annotation alternatives, exposed-data status, and distinct task/component/response denominators.
- Budget-reachability and same-set controls that separate answer-space impossibility, annotation omission, ranking sensitivity, and actual semantic loss; exact finite controls are checked by exhaustive enumeration.
- A new executed complete-score-support analysis that turns endpoint disagreements into sharp fixed-cohort contrast-sensitivity envelopes, with separate qualification-cohort transport bounds and explicit worst-case versus behavioral-estimate distinctions.
- A falsifiable, equal-information and fully charged comparison protocol against conservative replay, typed migration, and assessment/assembly methods; agent effects remain unestimated rather than inferred from grader diagnostics.
- A constrained, fully replayed merge operator exposes a score-sign reversal on the same task cohort, separating monotone evidence union, ranked truncation and exact-set grading without inventing model histories.

- An exact order/capacity benchmark with annotation-blind controls, permutation-orbit bounds, a complete sufficient-failure partition and a symmetry counterexample to interpreting local reversal repairs as aggregate improvement.
- An executed label-blind agreement audit with pairwise length-matched random controls, component-disjoint rule selection, restricted-class oracle accounting and a post-primary controlled common-priority counterexample.
- A new predeclared finite audit keeps the round-robin relative order within common and noncommon IDs and only promotes the common class. For a nested output prefix L, the first sufficient capacity is the minimum, over complete reference alternatives, of the largest reference-ID rank. Two operators therefore differ in sufficiency on at most one contiguous capacity interval for a fixed pair. This is an annotation-containment identity, not a semantic claim about agreeing natural skills.

## 26 Preserved v19: Introduction

Agent instructions increasingly live in reusable skill files with associated tools and resources. A deployment decision arises when two changes are useful separately: should both become available together, should integration be repaired, or should deployment fall back to an isolated branch? A negative score alone does not answer why integration failed. It could reflect lost evidence, different ordering under a ranked contract, an omitted valid annotation, changed tool rights, or additional context. Distinguishing these possibilities determines which intervention is worth deploying.

Consider an evidence task whose correct rationale contains three sentences. Two isolated configurations and their ancestor return these sentences followed by an irrelevant sentence; the joint configuration returns the same four IDs with the irrelevant one first. A first-three grader reports a strict joint failure, while full-set containment reports no change. If ranked evidence is the application contract, the failure is operationally meaningful. The identical full set nevertheless rules out the stronger explanation that the joint answer lost an annotated evidence sentence. A study that confuses these claims can prescribe the wrong repair.

Our question is therefore two-level. The behavioral target is whether independently qualified changes lose completion jointly. The completed empirical contribution is an audit of the measurement substrate required to investigate that target. This ordering follows established concerns about evidence evaluation [1,3,4] and construct validity [8,9], rather than claiming that measurement validity is newly discovered. The useful gap is operational: a claim about qualified composition needs a chain linking candidate history, held-out qualification, answer expressibility, grader behavior, and fresh deployment outcomes.

We ask three research questions. RQ1: do original valid annotation responses remain accepted, and is a complete answer expressible under the proposed budget? RQ2: which endpoint changes can occur while the full evidence set is fixed? RQ3: what additional histories, controls, and comparison rights are required before estimating a qualified-composition effect? RQ1 and RQ2 receive finite, executed answers; RQ3 receives an explicit prospective protocol. No observation in the completed audit qualifies a skill pair.

The v14 follow-up asks a sharper measurement question: given a task, a response budget and fixed arm outputs, how far could changing the endpoint move the composition contrast? Counting four particular order flips cannot answer that question, because those controls explore only selected responses. We therefore characterize all attainable simultaneous sufficient, first-three and exact scores within a declared response space. A sharp envelope over unrestricted response assignments can be calculated without inventing a single agent history. We also distinguish this same-cohort sensitivity from changing which candidate pairs qualify when the qualification grader changes.

Figure 1 separates these evidential stages. The completed lane contains 531 tasks, 309 evidence-bearing tasks, 548 annotation responses, and 4,186 constructed diagnostics. These are nested units, not independent integration events. The separate behavioral lane has zero authoring histories, zero eligible pairs, and a null effect. Our central practical result is that an incomplete grader and an unexamined response cap would already change eligibility and mechanism interpretations before any model comparison begins.

Figure 1 preserves the earlier measured-versus-prospective separation; Figure 4 adds the new response-space sensitivity analysis. The original 4,186 stored diagnostic rows are not relabeled as new model predictions, and the new abstract score cells are not qualifying events.

V15 asks which measurement patterns survive a concrete coupling between the isolated and merged responses. A union cannot independently choose an arbitrary response for each arm, and cannot invent a new sentence. By replaying that narrow operator we obtain a constructive control below

the v14 worst-case envelope: a score decrease can arise from a nonmonotone endpoint even when evidence is preserved, while a capacity-limited union can actually drop the second input's evidence. Both statements can be tested before collecting real skill histories.

V16 then asks whether a visible ranking repair survives a symmetric cohort comparison. Exact prefix-subset enumeration separates reference availability, capacity and order, while annotation-blind controls expose an endpoint tradeoff. The key negative control is that reversing every ordered pair has zero aggregate gain even when it repairs every selected priority failure. Figure 6 links that finite result to the still unexecuted behavioral study.

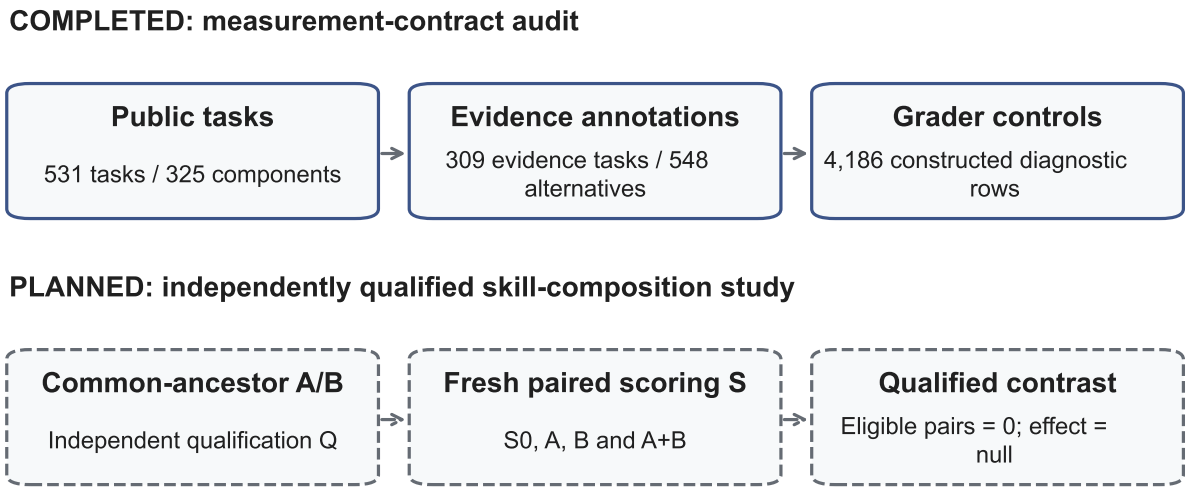



Figure 2: Completed measurement work and prospective behavioral inference are separate. Solid stages contain 531 tasks in 325 structural components, 309 evidence-bearing tasks in 204 components plus 222 insufficient tasks, 548 original annotation responses, and 4,186 constructed diagnostics. The dashed A/B-authoring, qualification-Q, and fresh-S lane has zero eligible histories and null effects. Counts are nested units; no arrow establishes observed composition collapse.

## 27 Preserved v19: Related Work

We organize related work by the decision it supports: what counts as evidence, what a benchmark score warrants, and what an integration policy actually does. This avoids treating a dataset, a testing philosophy, and an executable skill-management method as interchangeable performance baselines.

### 27.1 Preserved v19: Evidence and rationale evaluation

SciFact evaluates scientific claim–abstract relations with rationale annotations [1]. Its rationalized abstract endpoint uses containment of a complete rationale, whereas its sentence endpoint also penalizes excess predictions. FEVER likewise distinguishes label correctness from evidence-supported correctness and recognizes incomplete evidence annotation [3]. ERASER separates agreement with human rationales from faithfulness to a model's prediction [4]. These distinctions motivate separate endpoints; they do not justify naming exact annotation agreement semantic truth. Our sufficient-containment rule also differs from ERASER's confidence-based notion of rationale sufficiency.

The pinned SciFact implementation slices a supplied rationale list to three IDs for rationalized abstract correctness [2]. The inspected parser retains the supplied list rather than sorting it. We study that local list behavior explicitly. We do not reproduce corpus retrieval, native aggregate precision/recall/F1, or ERASER faithfulness. Original annotations are operational references, not independent domain adjudications.

### 27.2 Preserved v19: Behavioral testing and warranted inference

CheckList distinguishes focused functionality, invariance, and directional tests [5]. Our same-set controls apply this testing logic to deterministic graders, not to a model evaluated through CheckList. HELM motivates examining several desiderata in the same scenarios [6]; BIG-bench discusses sensitivity to task and metric specification [7]. Construct-validity analyses make the score-to-claim assumptions explicit [8,9]. We implement a narrow part of that agenda through executable acceptance and answer-space checks.

Underspecification warns that similar held-out predictive performance can coexist with different deployment behavior [11]. Datasheets motivate explicit composition, collection, and intended-use documentation [12], while The Benchmark Lottery motivates fixing task selection and weighting

rather than interpreting a convenient ranking as universal [13]. None proves that a particular skill pair interferes. Together they explain why our exposed annotation audit supports an engineering admission decision but not a behavioral population estimate.

### 27.3 Preserved v19: Skill evolution and integration controls

DSPy compiles modular language-model programs against an explicit metric, including metric-filtered traces [10]. SkillCommit emphasizes source-case replay and conservative scope management [14]. SkillSmith includes typed operations and required caller updates [15]. SkillCAT v2 uses assessment with an LLM router and an LLM assembler [16]. SkillEvoReg studies candidate-conditioned regression validation and repair [17]. These provide substantive comparison mechanisms, but optimize or condition on different objects; none is automatically an estimator of our qualified-pair contrast.

Signal or Noise separates skill instructions, matched irrelevant content, slices, and mounted resources [18]. Its byte-matched control motivates a length diagnostic, while actual tokenizer equality remains a separate requirement. SkillsBench's versioned task/environment/oracle/verifier packages offer a prospective workflow setting [19]. We inspect these primary sources and specify comparison rights; we execute none of their native systems, import none of their reported scores, and make no priority claim over their broader mechanisms.

## 28 Preserved v19: Problem Formulation and Method

An event fixes a common skill ancestor S0, two branch changes A and B, their associated resource graph, and a canonical joint configuration AB. Installation, routing, actual reading, and runtime assembly are distinct interventions. The primary behavioral treatment is joint availability under a declared deployment policy. A forced-activation test can diagnose a route, but cannot replace availability while retaining the same estimand.

### 28.1 Preserved v19: Qualification-conditioned target

Authors may use development information D but cannot inspect each other's change or auditor-only Q/S labels. An event qualifies only if both isolated branches satisfy a predeclared gain rule against the same ancestor on Q. Qualification must be based on observed task behavior, not an assigned role or a critic's approval. Proposal caps, duplicate attempts, authoring failures, and Q failures remain in a yield ledger. Selection on Q defines a conditional target; it does not measure the frequency of harmful merges among all proposed changes.

For a qualified event and task, p0, pA, pB, and pAB denote completion probabilities under fresh reset executions with fixed model identity, rights, contract, and accounting. The primary contrast compares joint deployment to an equal mixture of isolated branches. It does not select the best branch after inspecting final S. Weights over histories and source families must be declared before S; task-rich components otherwise receive unintended extra influence.

The interaction tau is different. The authored cell (S0,A,B,AB)=(0,1,1,1) gives tau=-1 but primary contrast zero: joint deployment is not worse than either isolated branch. A strict cell (1,1,1,0), a negative mean contrast, and a high-probability witness are also different claims. The current ledger contains no actual A/B histories, Q evaluations, or assigned S cells; all composition effects and witness incidence remain null.

$$d_{e,t} = p_{AB} - \frac{p_A + p_B}{2} \tag{21}$$

$$\theta_Q = E_{w[d_{e,t} \texttt{ mid } Q_e=1]} \tag{22}$$

$$\tau_{e,t} = p_{AB} - p_A - p_B + p_0 \tag{23}$$

### 28.2 Preserved v19: Evidence contracts

For an evidence-bearing task t, let its n sentence IDs be stable and let R1 through Rm be its original nonempty rationale sets. A response L is an ordered list of distinct in-range IDs. Every executed diagnostic receives the correct label from the auditor. This isolates evidence-contract behavior and deliberately excludes label prediction. Insufficient tasks use a separate explicit label and empty evidence in the local adapter.

Unlimited sufficiency accepts when a complete original rationale occurs anywhere in the response set. First-three applies containment only to the first three supplied IDs. Exact-annotated accepts when the full set equals an enumerated rationale. An intentionally incomplete first-annotation-only comparator retains only the first original rationale. The last comparator tests omission and is never proposed as an admissible deployed grader. Order invariance, tolerance of extra evidence, and preservation of alternatives are distinct properties.

The endpoint must follow the application. Containment can reward an answer with excessive unrelated material; exact matching can reject a valid unannotated explanation. Ranked evidence may be a legitimate retrieval requirement. We therefore audit contracts without promoting whichever endpoint makes a preferred composition story appear. The local cited-document task and its insufficient convention differ from native SciFact retrieval and aggregate scoring [1,2].

| Contract | Acceptance condition | What a rejection can mean |
|---|---|---|
| Unlimited sufficient | Some original rationale is contained in set(L) | No complete enumerated rationale in the full set |
| First three | Some original rationale is contained in set(L[:3]) | Evidence absent from the scored prefix; may still exist later |
| Exact annotated | set(L) equals one original rationale | Extra IDs or mismatch with enumerated alternatives |
| First annotation only | The first listed rationale is contained in set(L) | May reject a different original valid alternative |

Table 17: Contracts used in the executed measurement study, conditional on an auditor-correct label. This is not a ranking of universally better graders.

### 28.3 Preserved v19: Reachability, invariance, and exact controls

Proposition 1. Under an at-most-k sentence budget, an annotated sufficient answer exists if and only if the shortest original rationale has size at most k. Necessity follows because a response containing a complete rationale cannot be shorter than that rationale. Sufficiency follows by returning the shortest rationale itself. This is an answer-space condition, not a model success bound conditional on arbitrary generation behavior. A first-three cap consequently makes any task with only longer rationales unreachable under that endpoint.

Proposition 2. Full-set containment is invariant to permutations of L and monotone under adding IDs. Exact-annotated is permutation invariant but need not be monotone. First-three need not be permutation invariant. For a three-ID rationale R and an ID d outside every annotated rationale, compare [r1,r2,r3,d] with [d,r1,r2,r3]. Their full sets are identical and sufficient, but only the first passes first-three. Assigning the first list to S0/A/B and the second to AB constructs a strict score cell without losing any full-set evidence. This is a deterministic counterexample, not an eligible behavioral event.

A uniform k-subset control quantifies document-position structure without inventing predictions. With k=min(b,n), inclusion–exclusion over nonempty subsets J of distinct rationales gives the exact probability that the chosen set contains at least one rationale. Here uJ is the size of their union; impossible binomial terms are zero. Exact-annotated instead counts distinct size-k annotations divided by the total number of size-k subsets. Exact rational arithmetic and exhaustive checks avoid Monte Carlo uncertainty. These elementary results are useful contract checks, not claims of mathematical priority.

$$\text{reachable}_t(k) = I\left[\min_j |R_{t,j}| \le k\right] \tag{24}$$

$$q_{t,k} = \sum_{J \neq \varnothing} (-1)^{|J|+1} \frac{\binom{n_t - u_J}{k - u_J}}{\binom{n_t}{k}} \tag{25}$$

### 28.4 Preserved v19: When endpoint disagreement changes a composition contrast

Endpoint disagreement alone does not determine the direction of a composition effect. Hold the qualified cohort Q, task weights, and arm outputs fixed, and let eta_z be the expected difference between two binary graders g and h on arm z. Linearity gives the contrast decomposition below. If every arm has the same discrepancy, it cancels. A task that is unreachable under one contract for all arms therefore imposes a ceiling but need not create a negative joint-minus-isolated contrast. Its effect on qualification or weighting is a separate question.

If epsilon_z is the weighted probability that the graders disagree on arm z within that same fixed Q cohort, the triangle inequality yields the displayed sensitivity bound. These are conditional algebraic statements, not numerical estimates from our oracle diagnostics. We have no arm-specific model outputs, so neither eta nor epsilon is observed for a qualified composition population. In particular, 239 omitted-alternative rejections and four constructed order flips cannot be substituted into the bound as model disagreement probabilities.

Changing the grader used for Q may also change which histories qualify. Then even the cohort differs, and the fixed-Q identity cannot transport one reported effect to the other without additional assumptions or observations. A future study should distinguish rescoring the same frozen S outputs

from rerunning qualification under a different contract. The former diagnoses measurement sensitivity; the latter changes the conditional study population. This distinction prevents treating a convenient endpoint switch as either a repaired mechanism or evidence of general collapse.

$$\theta_Q^g - \theta_Q^h = E_w\left[\eta_{AB} - \tfrac{\eta_A+\eta_B}{2} \texttt{ mid } Q{=}1\right] \tag{26}$$

$$\left|\theta_Q^g - \theta_Q^h\right| \le \varepsilon_{AB} + \frac{\varepsilon_A + \varepsilon_B}{2} \tag{27}$$

### 28.5 Preserved v19: Complete score support and sharp fixed-cohort envelopes

For this follow-up, an admissible response is an ordered list of distinct, in-range sentence IDs of length at most k, including the empty list. Labels are auditor-correct. Let U, T and E denote sufficient, first-three and exact-annotated scoring. Both T and E imply U, so only five binary patterns can occur: (0,0,0), (1,0,0), (1,0,1), (1,1,0) and (1,1,1). The response-support algorithm records a minimum-length witness for every attainable pattern, not merely patterns observed in a few prefix controls.

Every current task has a sentence ID outside the union of all enumerated rationale sets. This is checked rather than assumed for an arbitrary future benchmark. Exact responses are the original rationale sets with each relevant three-ID prefix. A short rationale followed by that outside-union ID supplies a sufficient/first-three but nonexact witness. To complete the remaining sufficient-but-not-first-three and nonexact pattern, enumerate unsuccessful three-ID prefixes and complete an original rationale after them. If their union is exact, append the outside-union ID. The appendix proves that any longer admissible witness can be reduced to one of these constructions without increasing its length.

For endpoints g and h and a fixed task t, let the discrepancy be g(L)-h(L). Its minimum and maximum over admissible responses define a taskwise range. With nonnegative fixed task weights summing to one, the sharp symmetric envelope for the change in joint-minus-mean-isolated contrast is the weighted mean of those ranges. Choose the joint response at the maximum discrepancy and both isolated responses at the minimum to attain the upper endpoint; reverse them for the lower endpoint. Sharpness is relative to unrestricted deterministic response assignments independently chosen for each arm and task. It does not assert that any qualified skill or learning procedure can generate the extremizer.

This support envelope refines the earlier disagreement-probability inequality when only endpoint structure is known. For nested binary graders, discrepancies have one sign and the taskwise range is at most one, rather than the unconstrained two. First-three and exact are not generally nested. In a five-ID authored example with rationales (0) and (0,1,2,3), responses (0,4) and (1,2,3,0) have discrepancies +1 and -1. Assigning them to opposite arms attains a contrast change of two. No task in the current data has that two-sided pattern support. Dataset-specific absence must not become a universal nesting claim.

$$m_t^{gh}(k) = \min_{L \in \mathcal{L}_t(k)} \left(g(L) - h(L)\right) \tag{28}$$

$$M_t^{gh}(k) = \max_{L \in \mathcal{L}_t(k)} \left(g(L) - h(L)\right) \tag{29}$$

$$R_w^{gh}(k) = \sum_t w_t \left(M_t^{gh}(k) - m_t^{gh}(k)\right) \tag{30}$$

$$-R_w^{gh}(k) \le \theta_w^g - \theta_w^h \le R_w^{gh}(k) \tag{31}$$

Assert stable sentence IDs, nonempty original rationales, and an ID outside their union.
Add the empty response; enumerate exact rationales and their relevant three-ID prefix subsets.
Add each length-at-most-three rationale followed by the outside-union ID.
Enumerate unsuccessful three-ID prefixes; complete each original rationale after the prefix.
When that union equals an original rationale, append the outside-union ID if budget permits.
Keep a minimum-length witness for each simultaneous score pattern.
At each budget, derive discrepancy extrema and verify both signed contrast-envelope endpoints.
Report actual behavioral effects as null until independent qualified histories exist.

Algorithm 5: Executed finite support certificate under the stated response contract; no model or qualification history is synthesized.

### 28.6 Preserved v19: Changing qualification is a separate transport problem

A qualification grader can change the target cohort as well as final-output scores. To make the distinction executable, we exhaust an illustrative one-task binary gate: both isolated responses must strictly outperform the same ancestor. Every ancestor/A/B score cell uses a feasible response pattern.

The resulting counts identify tasks on which a gate change is possible, not the frequency of changes among real proposals and not the study's eventual multi-task qualification rule. Even when g never scores a response below h, the gate can move in either direction: g may upgrade a weak isolated branch or upgrade the ancestor and eliminate its measured gain.

For a common original cohort measure and unchanged history weights, let q_g and q_h be the masses admitted by two nonempty gates and r their intersection mass. The total variation distance between the two normalized selected distributions is one minus r divided by the larger admission mass. A mean-isolated contrast lies in [-1,1], so changing only that selected distribution can move its expectation by at most twice this distance. This is a finite algebraic transport bound, not an assumption that the two gates admit equal histories.

Decompose a full grader-and-gate switch through the h score on the g cohort. The first term is same-cohort rescoring and admits the support envelope; the second is cohort transport and admits the total-variation bound. Their sum can be capped by two because the two reported contrasts lie in [-1,1]. This comparison requires fixed candidates, common base weights, and enough retained final-output support to define the bridge quantity. If access rights, authoring behavior or final executions change, those are additional interventions. If either qualification cohort is empty, its conditional effect and this transport comparison are undefined and remain null. Current gate memberships, admission masses and overlap are unobserved; no numerical transport correction is applied to a behavioral effect.

$$V = 1 - \frac{r}{\max\left(q_g, q_h\right)} \tag{32}$$

$$\left| E_{Q_g}[d^h] - E_{Q_h}[d^h] \right| \leq 2V \tag{33}$$

$$\left| \theta^g_{Q_g} - \theta^h_{Q_h} \right| \leq \min\left(2, R^{gh}_{Q_g} + 2V\right) \tag{34}$$

## 29 Preserved v19: Experimental Setup

The completed experiments operate on public dataset records and authored oracle-label response lists. The earlier audit establishes source reconstruction and basic grader behavior; the follow-up measurement run adds budget, alternative, same-set, and combinatorial diagnostics. The present revision adds analysis and inspected literature, not another predictive experiment.

### 29.1 Preserved v19: Sources, splitting, and exposure

The preserved raw snapshot contains 809 training claims, 300 development claims, and 5,183 abstracts. An independent breadth-first reconstruction checks the prepared adapter against raw bytes, including original labels and rationale alternatives. Excluding 278 training claims linked to development sources leaves 531 local tasks in 325 structural components. Of these, 309 evidence-bearing tasks occupy 204 components; 222 are insufficient tasks. These are local adapter counts, not a substituted official benchmark population.

Recorded cross-split component, selected-document, and normalized-claim overlaps are zero under the reconstruction rules. Structural separation is narrower than semantic independence or pretraining-contamination freedom. All 531 selected labels and annotations were already exposed during development audit. None is a blind final set for a newly authored skill. The exposure ledger follows the documentation rationale of [12], but no documentation artifact itself guarantees independence.

The rationale family has 548 original annotation records across 309 evidence tasks. Task-level shortest-rationale sizes are one on 290 tasks, two on 15, three on three, and four on one. Across all annotation records, the corresponding counts are 510, 33, four, and one. Alternatives share tasks and sources; treating 548 responses as independent tasks would change the denominator without adding independent evidence.

### 29.2 Preserved v19: Diagnostic construction and validation

For each original annotation we construct its ordered IDs with the correct label. For each budget from one through ten we also return a document prefix of at most that many IDs. Reachability is computed independently of this prefix policy. Finally, for every annotation, we choose one sentence outside the union of its task's annotated rationales and place the same ID before versus after the annotation. All 309 evidence tasks support this construction; no task is silently excluded. The full set, label, and distractor identity are fixed within a pair.

The saved run contains 4,186 diagnostic rows. Basic scorer checks include original annotation/empty responses, wrong labels, evidence deletions, and malformed outputs from the preceding audit. Follow-

up mathematical checks cover overlapping and duplicate rationale families as well as boundary budgets. Saved traces are reaggregated and their input/output byte bindings checked; all 66,431 real-data size-three subsets are exhaustively enumerated to verify the analytic control. This establishes specified software relationships, not label correctness or independent scientific replication.

No model, router, assembler, or tool-using agent produces these responses. The execution makes zero model calls and zero network requests. Later literature HTML retrieval belongs to this revision's source review, not to the offline experiment. We preserve the executed plan, script snapshot, results, row traces, and receipts, including an explicit supplement for a prepared-manifest binding omitted from the original follow-up receipt.

### 29.3 Preserved v19: New finite support analysis and checks

One new offline run freezes the analysis code, plan, task measurements, prior diagnostics, prior result and prior byte receipt before support analysis. It uses all 309 evidence tasks in 204 structural components, with all 548 rationale alternatives. All labels were already development-exposed. The empty response is an explicit failure response in the at-most-budget contract, rather than a new semantic annotation. This follows the earlier correct-label endpoint definitions; it is not native SciFact aggregate evaluation.

We compare certificate support against complete ordered-response enumeration for all 127 nonempty rationale families on three evidence IDs plus a fourth outside-union ID: 16,256 score checks. All 152 real tasks with at most eight sentence IDs receive independent response-set/prefix enumeration at budgets one through four, giving 608 task-budget support comparisons. The preserved 4,186 diagnostic rows expand to 4,734 response instances because order-pair rows contain two responses; all three endpoint scores are independently recalculated. The main analysis checks 460,020 feasible three-arm score cells and attains both signed envelope extrema. These are dependent software checks on a fixed collection, not independent model trials.

A separate verifier checks byte bindings, rescores 1,222 stored minimum-length support witnesses, and recomputes all 30 aggregate rows. Because the small-family universe cannot contain a four-sentence rationale, it separately brute-forces the actual unique four-sentence task at budgets three, four and five. Five adversarial rationale families add 3,250 ordered-response checks, including mixed short/long annotations with two-sided first-three/exact discrepancy. All checks pass. The main finite analysis took approximately 0.88 seconds in this local execution; this is bookkeeping, not a controlled efficiency comparison.

## 30 Preserved v19: Results and Analysis

Results answer admission and interpretation questions on a fixed exposed corpus. Exact counts require no sampling interval for this corpus; they do not estimate a population composition rate. We retain endpoint-specific denominators and distinguish existential reachability from achieved diagnostic scores.

### 30.1 Preserved v19: Budget changes expressibility before behavior

Figure 2 plots the answer-space ceiling and the constructed prefix policy on the same 309 tasks. Budgets one, two, three, and four permit a complete annotated answer on 290, 305, 308, and 309 tasks respectively. The unique four-sentence-only annotation remains preserved; a three-ID endpoint cannot accept it even with perfect sentence selection. A historical paper description of rationale length does not replace checking the actual raw snapshot and adapter.

Document-prefix sufficient passes rise from 17 at budget one to 76 at three, 119 at four, and 251 at ten. First-three passes plateau at 76 after budget three. Exact-annotated prefix passes are 17, three, and zero at budgets one, two, and three; they remain zero through ten. These curves measure different contracts, not a model becoming worse at verification as its budget increases. Under exact matching, extra predicted IDs can change a previously exact set into a rejection by definition.

This changes an immediate design decision: either preserve the ranking cap and explicitly account for unreachable original cases, or define a justified adapted task before testing new behavior. Silently discarding the four-sentence case or increasing the cap after observing a desired effect would change the target. We make neither change and promote no endpoint.

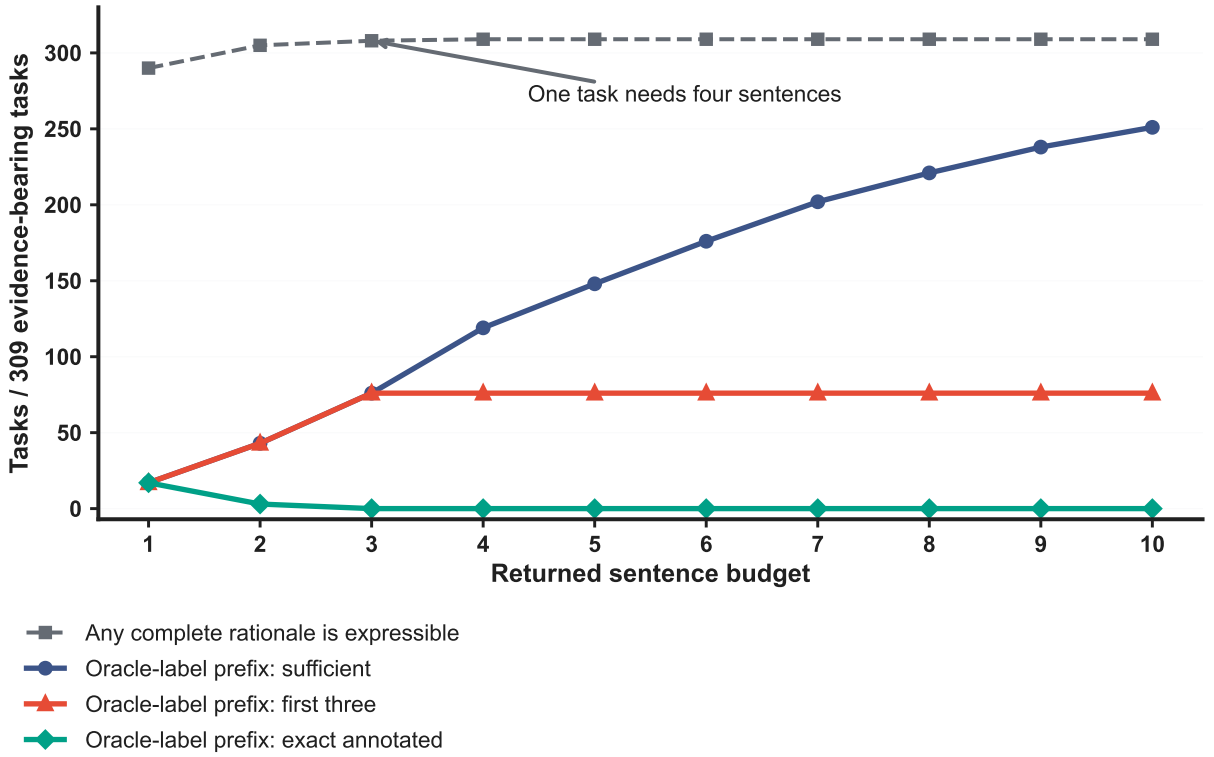



Figure 3: Sentence budgets 1–10 on the same 309 evidence-bearing tasks. Annotation reachability is an existential answer-space ceiling, while prefix sufficient/first-three/exact curves score constructed document-prefix responses with auditor-correct labels. At k=3, 308 tasks are reachable but only 76 prefix responses pass sufficient containment; the first-three curve then plateaus. These are not LLM accuracies.

### 30.2 Preserved v19: Alternative preservation and same-set order effects

Every original annotation response passes unlimited sufficiency and exact-annotated: 548/548 each. First-three accepts 547/548. The incomplete first-annotation-only comparator accepts 309/548, losing 239 original valid annotation responses, or 43.61% of these response units. This is not a 43.61-point model gain: the same task can contribute several alternatives, and no model chose among them. It is a concrete reason to reject that incomplete grader for the stated annotation-preserving contract.

Figure 3 compares original responses with the same one-distractor set in two orders. Unlimited sufficiency accepts 548/548 in both orders. First-three accepts 543/548 when the distractor is prepended and 547/548 when appended, producing four paired flips. Exact-annotated rejects every distractor response in both orders because the chosen ID lies outside every rationale. The incomplete comparator accepts 309 in either order. Four flips are an observed count among constructed annotation pairs, not the natural incidence of harmful merges.

The control rules out full-set evidence loss as an explanation for these particular flips. It does not invalidate ranking as a task requirement, nor does it establish that model attention, instruction conflict, or context length caused a behavioral change. Those require model-produced outputs and matched interventions. An integration policy whose only benefit is moving evidence into a required prefix may be useful, but its claim should name ranking repair.

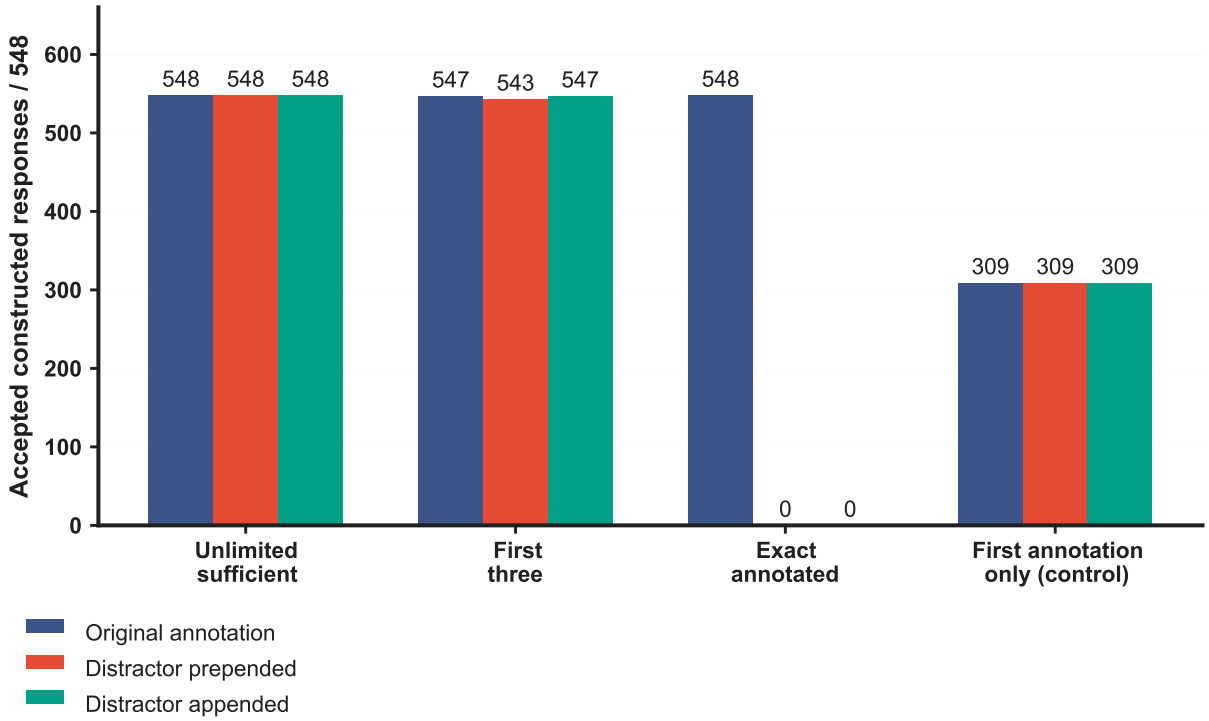



Figure 4: Pass counts for 548 original annotation responses, nested in 309 tasks, under four contracts. Original/prepend/append counts are 548/548/548 for unlimited sufficiency, 547/543/547 for first-three, 548/0/0 for exact-annotated, and 309/309/309 for the deliberately incomplete first-annotation-only control. Prepend/append hold the full set fixed; their four first-three flips are constructed diagnostics, not composition incidence.

### 30.3 Preserved v19: Finite controls and the missing behavioral denominator

At size three, uniform-subset sufficient containment has exact equal-task expectation 0.490307, compared with document-prefix containment 76/309, approximately 0.245955. This diagnoses sentence-position structure under correct labels; it does not show a random agent outperforming a verification model. Equal-component averaging yields 0.496126, numerically different because tasks and components induce different targets. Reporting one as the other would hide a weighting choice.

The authored formula passes 165 exhaustive small-family comparisons and five additional contract-check groups. Independent saved-trace validation reaggregates all 4,186 rows, checks 25 bound input/output files, and confirms the real-data three-subset enumeration. The elementary propositions and diagnostic counts are consequently supported as finite software/data statements, conditional on the annotations and definitions.

The behavioral funnel remains empty: actual A/B histories=0, Q-evaluated pairs=0, eligible pairs=0, and assigned S cells=0. Theta_Q, interaction, and strict-witness rate are null. The nulls mean missing behavioral evidence, not zero harm. Neither successful software tests nor a model critic's agreement can populate those denominators.

| Evidence object | Observed denominator | Supported interpretation |
|---|---|---|
| Prepared public-data tasks | 531 tasks / 325 components | Reconstructed structural subset; already exposed |
| Evidence annotation diagnostics | 548 responses / 309 tasks | Contract acceptance and paired order controls |
| Saved diagnostic traces | 4,186 rows | Reproducible constructed-response calculations |
| Qualified composition | 0 histories / 0 eligible pairs | Effects null; no behavioral incidence estimate |

Table 18: Different denominators answer different questions. Response rows and exhaustive subsets do not create independent skill histories.

### 30.4 Preserved v19: Response-space sensitivity is not observed disagreement incidence

Figure 4 reports the new support-envelope radii. At budgets one through three, sufficient and first-three give identical scores for every admissible response, so their fixed-cohort contrast difference is exactly zero. At budget four, 296 of 309 tasks support both zero and positive sufficient-minus-first-three discrepancy. Their equal-task radius is 296/309, approximately 0.957929, and it remains unchanged through budget ten. Equal-component weighting yields approximately 0.967320. The 13 remaining tasks have no such score variation in this response space. A large worst-case radius means

annotation and endpoint structure alone cannot guarantee sign robustness; it does not mean models usually approach this extremum.

Sufficient versus exact has equal-task radii zero at budget one, 290/309 at two, 305/309 at three, 308/309 at four, and one from five onward. The unique four-sentence rationale is exact at four IDs and becomes a sufficient but nonexact response when an extra ID is permitted at five. First-three versus exact has radius 305/309 at budget three and one from four onward. All current taskwise radii are zero or one, although the adversarial family demonstrates that radius two is possible on a different annotation structure.

These results explain why the earlier four particular one-distractor order flips cannot be used as an overall error probability or sensitivity bound. Those controls expose four changes within their selected paired responses; complete response support allows other scored prefixes with no annotated rationale followed by a valid rationale. Conversely, zero sufficient/first-three sensitivity at budget three does not validate that budget: the earlier audit shows one original task whose four-sentence-only answer is unreachable for both graders. Agreement between endpoints can coexist with a common answer-space failure.

The envelope is calculated on the exposed development collection with declared weights. It is not R_Q for an unknown qualified population and cannot be substituted for theta_Q. A future same-output score rescoring can use its own arm-specific disagreements, which may cancel and be much smaller than the worst-case support radius. A future negative contrast is sign-robust only if its uncertainty and declared endpoint-sensitivity allowance support that conclusion. No such behavioral estimate or uncertainty interval exists in the current study.

| Budget | Sufficient / first-three radius | Sufficient / exact radius | First-three / exact radius |
|---|---|---|---|
| 1 | 0 | 0 | 0 |
| 2 | 0 | 290/309 | 290/309 |
| 3 | 0 | 305/309 | 305/309 |
| 4 | 296/309 | 308/309 | 1 |
| 5 | 296/309 | 1 | 1 |

Table 19: Sharp symmetric response-space sensitivity radii, equal weights over 309 evidence tasks. The contrast change lies between minus and plus the radius under unrestricted arm-response assignments. No entry is a model accuracy or observed composition effect.

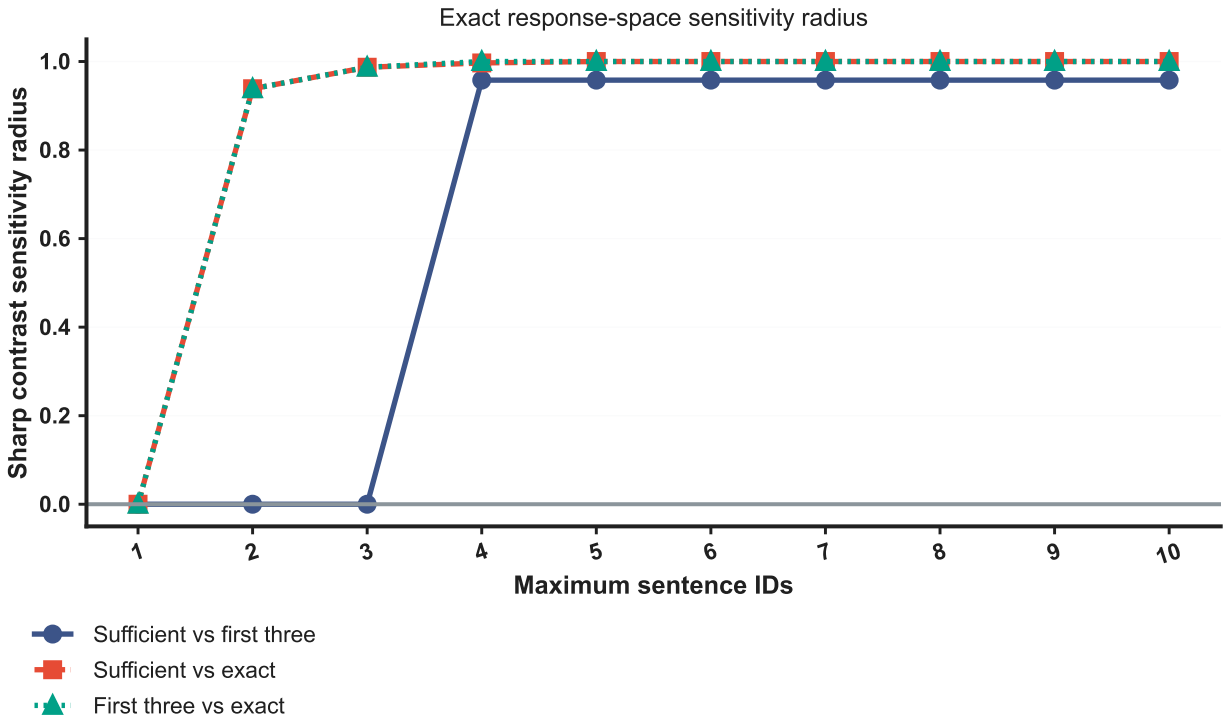



Figure 5: New v14 finite response-space sensitivity analysis on the same 309 development-exposed evidence tasks. Each curve is the sharp equal-task radius of endpoint-induced change in joint-minus-mean-isolated contrast, conditional on correct labels, at-most-k distinct sentence IDs, fixed task weights and unrestricted per-arm response assignment. Sufficient/first-three is zero through k=3 and 296/309 from k=4. Sufficient/exact reaches one at k=5; first-three/exact reaches one at k=4. Curves are worst-case envelopes, not model effects or error frequencies.

### 30.5 Preserved v19: Nested graders do not make qualification changes one-sided

At budget four, the illustrative single-task gate can admit under sufficient but not first-three on 296 tasks; it can admit under first-three but not sufficient on 295 tasks. These are overlapping task-level possibility counts, not observed gate outcomes or two groups of candidate histories. The asymmetry comes from the four-sentence-only task: first-three never accepts an admissible answer there, so it cannot qualify an improvement under that gate. Thus responsewise leniency does not imply selectionwise leniency. Any study that changes the Q grader must retain admission and overlap denominators, not merely rescore final S on the original selected cohort.

For the real target, no authoring histories, Q evaluations, eligible pairs or assigned S cells exist. The actual Q overlap, behavioral endpoint-disagreement frequencies and composition effects stay null. The useful new result is the exact separation between what endpoint structure can certify and what only a candidate-history experiment can identify.

## 31 Preserved v19: Constrained union-replay protocol

The unrestricted v14 response-space radius permits any admissible deterministic output per arm. It does not describe a learned merger or impose a realizability relation between A, B and AB. We now impose one explicit relation: AB is the ordered deduplicated union of A followed by previously unseen IDs from B. A second operator returns only that union's first three IDs. Neither is a model or a skill implementation; they are mechanistic controls for what a measurement can call a failure.

We freeze the prior task_measurements.json and diagnostic_rows.jsonl before execution. Every explicit ids, append_ids and prepend_ids field is inspected. The 4,734 explicit responses include oracle original annotations and deliberately constructed diagnostics. Within each task we deduplicate ordered responses and retain only responses of length at most three. This yields 2,494 unique input responses on all 309 evidence-bearing tasks in 204 structural components; no task has an empty retained bank. The restriction deliberately excludes a four-sentence-only original rationale as an isolated full answer. It is an expressibility constraint, not a successful qualification decision.

All ordered pairs from each task's retained bank are replayed, including A=B and both A,B versus B,A. There are 22,794 pairs and 45,588 operator applications. Ordered pairs are necessary because a ranked-capacity merger prioritizes A. Identical pairs remain because idempotence is part of the declared control. Pair multiplicities are diagnostic design choices, not sampled model-use frequencies. The pair-pooled denominator and the equal-task estimand are therefore reported separately.

For each endpoint we compute merged score minus the arithmetic mean of the two isolated scores. The primary summary averages first over every task's ordered pairs and then equally over 309 tasks. An equal-component result first averages task summaries inside each of the 204 components. Pair pooling weights tasks with a larger retained response bank more heavily and is a sensitivity summary rather than a replacement for the primary aggregation.

Input labels are auditor-correct throughout. Each isolated input has at most three sentence IDs; untruncated union can have six, whereas prefix3 has at most three. Thus an untruncated gain is not evidence of superiority at identical output cost. The analysis fixes task identity and label correctness, not realized tokens, model inference or context usage. No Q score or candidate history is generated.

$$M_{\cup}(A, B) = \text{ordered-deduplicate}(A \text{ then } B) \tag{35}$$

$$M_3(A, B) = \text{first-three}(M_{\cup}(A, B)) \tag{36}$$

$$d_g(A, B) = g(M(A, B)) - \frac{g(A) + g(B)}{2} \tag{37}$$

### 31.1 Preserved v19: Monotone evidence can receive a negative exact-match contrast

An untruncated union contains every evidence ID returned by either input. Under the sufficient endpoint, any accepted input rationale is therefore still present. We verify zero losses relative to either input on every pair. Nevertheless exact matching is nonmonotone: when A and B contain different individually valid rationale alternatives, their union may equal no original annotation set. Original alternatives [0] and [1] provide the minimal counterexample: both isolated answers are exact, while union [0,1] is sufficient but nonexact.

The executed equal-task union contrast is +0.199744 for sufficient, +0.136890 for first-three and -0.179878 for exact. Equal-component values are +0.198667, +0.136938 and -0.178035, so the sign reversal does not disappear under that declared regrouping. Pair-pooled values are +0.185838, +0.129815 and -0.206414. These are three different weighting summaries of the same diagnostic objects, not independent replications.

Exact grading produces 700 ordered pairs whose two isolated responses pass but merged union fails; sufficiency produces zero. Exact union also falls below at least one input in 8,718 pairs. A negative exact contrast is operationally real if exact annotation equality is the task contract. It is insufficient evidence that evidence content was lost, since every merged output is verified to contain both input sets. We do not redefine exact failure away; we restrict its mechanism interpretation.

The union never beats both isolated responses under sufficiency in this retained bank. Its positive contrast comes from cases with one successful and one unsuccessful input, not observed complementary partial rationales becoming a full answer. This negative coverage fact matters: the existing diagnostic construction does not span every behavior a real merger might exhibit. No frequency in this bank estimates the probability of such complementarity in model outputs.

| Operator / grader | Equal-task contrast | Both inputs pass, merge fails | Below either input |
|---|---|---|---|
| union / sufficient | +0.199744 | 0 | 0 |
| union / first_three | +0.136890 | 0 | 1277 |
| union / exact | -0.179878 | 700 | 8718 |
| prefix3 / sufficient | +0.136890 | 0 | 1277 |
| prefix3 / first_three | +0.136890 | 0 | 1277 |
| prefix3 / exact | -0.177863 | 699 | 8704 |

Table 20: Finite deterministic replay, 22,794 ordered pairs per operator/endpoint over the same 309 tasks. Pair counts are not population event frequencies; the contrast uses equal-task weights.

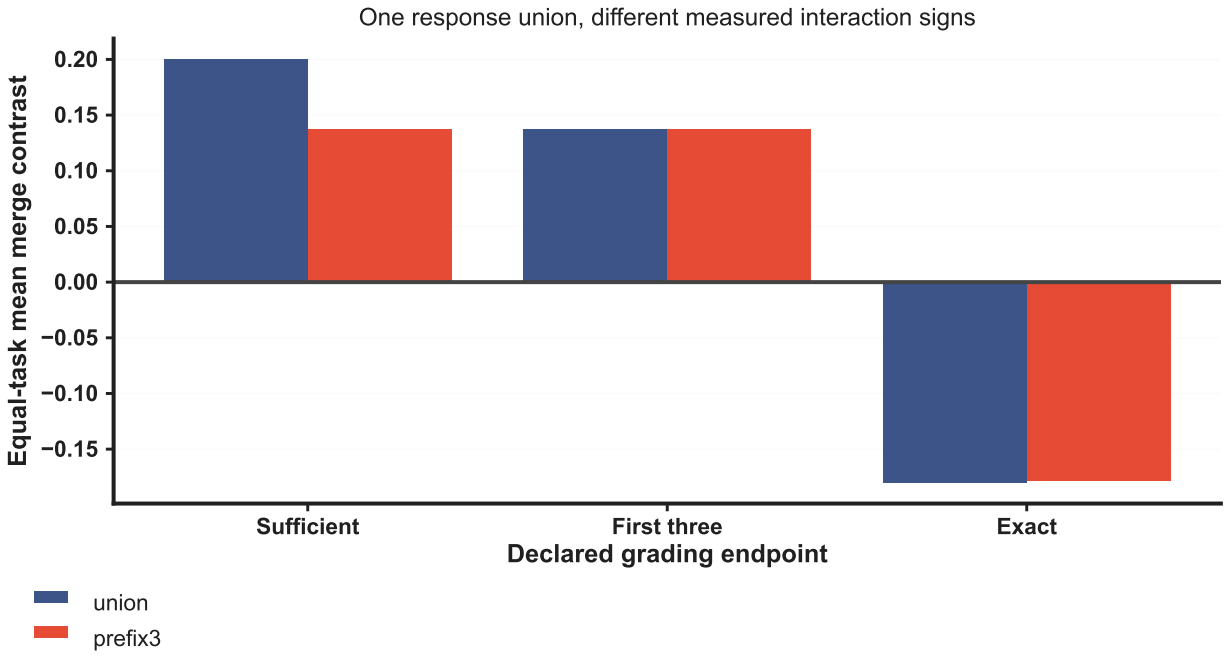



Figure 6: New v15 deterministic response coupling. Ordered union has positive sufficient and negative exact-grading contrast on the same fixed task bank. Prefix3 exposes ranked capacity loss. All values condition on correct labels and constructed responses; no observed model interaction or independent sample claim.

### 31.2 Preserved v19: Capacity and input order are distinct interventions

Prefix3 sufficiency exactly equals first-three union scoring on every pair, producing equal-task contrast +0.136890. Compared with untruncated sufficient union, capacity reduces that diagnostic contrast by 0.062854. This is an operator comparison over identical inputs, not a confidence interval or an inferred natural loss rate. The explicit full outputs distinguish actual sentence omission under truncation from mere endpoint rejection under exact grading.

There are 1,277 ordered pairs for which first-three union, equivalently sufficient prefix3, scores below at least one input. No pair loses when both inputs pass, because the A-prioritizing operator preserves all of A's at-most-three IDs. This asymmetry is built into the operator and explains why a both-pass criterion alone misses loss of an individually useful B when A fails. Both A,B and B,A are included so that this chosen prioritization remains visible.

One counterexample has A=[0,1,2], B=[3], with original rationale [3]. Ordered union preserves the rationale, its first-three score fails and a prefix3 output actually omits ID3. Reversing the inputs places ID3 first and avoids that particular capacity failure. In contrast, reordering a union containing

two distinct exact alternatives cannot make its full set exact. A repair aimed at ranking and a repair aimed at the acceptance contract therefore address different causal constructions.

We independently recompute all 136,764 pair-by-endpoint/operator cells without importing the author scorer, checking all 1,854 task summaries and both-pass rejection counts. Idempotence, no invented IDs, distinct-ID caps, sufficient monotonicity and the exact/truncation counterexamples are executable assertions. These calculations validate the finite replay implementation on exposed annotations; they do not validate an authoring system or establish independent data.

The next behavioral experiment must keep the union and prefix controls as declared baselines, preserve every initiated authoring history, qualify A and B using Q without S feedback and evaluate all arms on S with explicit costs. Until those histories exist, actual_A_B_histories=0, eligible_pair_n=0 and the qualified interaction effect remains null. A response-bank replay cannot silently fill that denominator.

## 32 Preserved v19: Exact order and capacity diagnostics

The v15 prefix counterexample leaves a tempting but incomplete prescription: reverse the two inputs. That repairs a chosen example, but what happens when every ordered pair is retained? We make this intervention explicit before claiming a better integration strategy. The research question now holds the available sentence set fixed and distinguishes a missing reference rationale, a rationale too large for the declared output, and a rationale that could fit but is excluded by one ordering.

The new plan and code are frozen with the exact earlier task measurements, diagnostic rows and v15 result before execution. We retain all 2,494 distinct at-most-three-ID input responses, all 309 evidence tasks, all 204 structural components and all 22,794 ordered pairs, including identical pairs. This continuation reuses exposed oracle-authored diagnostics. It makes 638,232 deterministic output replays across four controls and seven capacities; these are not new model predictions or independent behavioral trials.

Left union, right union, ascending document IDs and round-robin interleaving see only A and B. Each deduplicates the same union U and emits its first min(c,|U|) distinct IDs for capacities c=0 through6. Round robin interleaves A0,B0,A1,B1 and so on before deduplication; document order uses numeric sentence position rather than a relevance oracle. All controls therefore share the same available information and output-ID count for a given pair and capacity. Equal IDs do not imply equal tokens, inference cost, content quality or runtime.

For a uniform permutation of U, every size-k prefix subset is equally likely, where k=min(c,|U|). We enumerate these subsets rather than sample seeds. Sufficient and exact expectations use k; the first-three endpoint uses min(c,3,|U|). For every endpoint we also calculate the smallest and largest score over the orbit. Those extrema use annotations and belong only to the auditor. No policy receives the successful prefix or a gold-derived ordering.

The primary estimand is the equal-task mean score, averaging declared pairs within each task first. Equal-component and pair-pooled means remain separate sensitivities. We report score levels rather than silently changing the isolated-arm capacity in a new interaction contrast. The new code separately reproduces every one of the six v15 contrasts under its original alone-response contract. No endpoint is renamed as ERASER confidence-based sufficiency or an official SciFact leaderboard score [1,2,4].

$$k = \min(c, |U|) \tag{38}$$

$$P_{\text{suff}}(U, c) = \frac{\text{successful size-k subsets of U}}{\text{all size-k subsets of U}} \tag{39}$$

$$P_{\text{exact}}(U, c) = \frac{\text{size-k reference alternatives contained in U}}{\text{all size-k subsets of U}} \tag{40}$$

### 32.1 Preserved v19: A disjoint failure diagnosis and its reference boundary

For a failed sufficient left-prefix score, first ask whether any supplied reference rationale is contained in the full union. If none is, classify information_unavailable. Otherwise ask whether at least one contained rationale has size at most min(c,|U|). If none fits, classify capacity_unreachable. Every remaining failure is order_contingent. A success is the fourth class. This sequence makes the failure categories disjoint and exhaustive without claiming access to latent semantic correctness.

At capacity three, left-union sufficient failures divide into 2,255 pairs with no reference rationale anywhere in the available union, zero with a present but unfit rationale, and 1,277 with a fitting rationale excluded by the chosen order. These three classes plus 19,262 successful pairs exhaust all

22,794 pairs. The partition is conditional on the supplied reference alternatives: unavailability here is not a proof that the source text lacks other semantically valid evidence. A failed score therefore determines different next checks depending on whether information, capacity or ordering is implicated.

At capacity one the same partition contains 2,255 information-unavailable, 871 capacity-unreachable and 9,630 order-contingent failures, plus 10,038 passes. At capacity two only 28 capacity-unreachable cases remain. At capacities four and five there are still 67 and three left-priority ordering failures. Full capacity six removes every ordering failure but leaves the same 2,255 unavailable-reference cases. Thus neither adding a slot nor reranking can supply a reference rationale absent from both inputs.

The cap-three permutation orbit contains 2,574 pairs whose sufficient score changes under some ordering, whereas exchanging only source priority gives 2,554 sufficient disagreements. Merely testing two source orders is therefore not a complete permutation-invariance test. Exact matching has only 28 order-contingent pairs at that capacity and becomes order-invariant from capacity four in this bank. It can still fail because the final set contains extras; absence of ordering sensitivity is not acceptance.

| Capacity | Pass | No reference in U | Cannot fit | Order contingent |
|---|---|---|---|---|
| 0 | 0 | 2255 | 20539 | 0 |
| 1 | 10038 | 2255 | 871 | 9630 |
| 2 | 17249 | 2255 | 28 | 3262 |
| 3 | 19262 | 2255 | 0 | 1277 |
| 4 | 20472 | 2255 | 0 | 67 |
| 5 | 20536 | 2255 | 0 | 3 |
| 6 | 20539 | 2255 | 0 | 0 |

Table 21: Left-union sufficient partition on 22,794 constructed pairs at every capacity. Classes describe supplied-reference availability and the declared operator, not semantic truth, sampled model errors or independent events.

### 32.2 Preserved v19: A local reversal repair has zero symmetric-cohort gain

All 1,277 v15 losses below a passing input at capacity three are repaired by putting B first. That statement conditions on a failure selected after its score is known. Across the complete ordered cohort, swapping A and B is a bijection: the right-priority output for (A,B) equals the left-priority output for (B,A). Their sums are therefore exactly equal within every task, endpoint and capacity. Equal-task and equal-component aggregation preserve equality. A visible repair gallery cannot establish an aggregate advantage if its symmetric regressions are omitted.

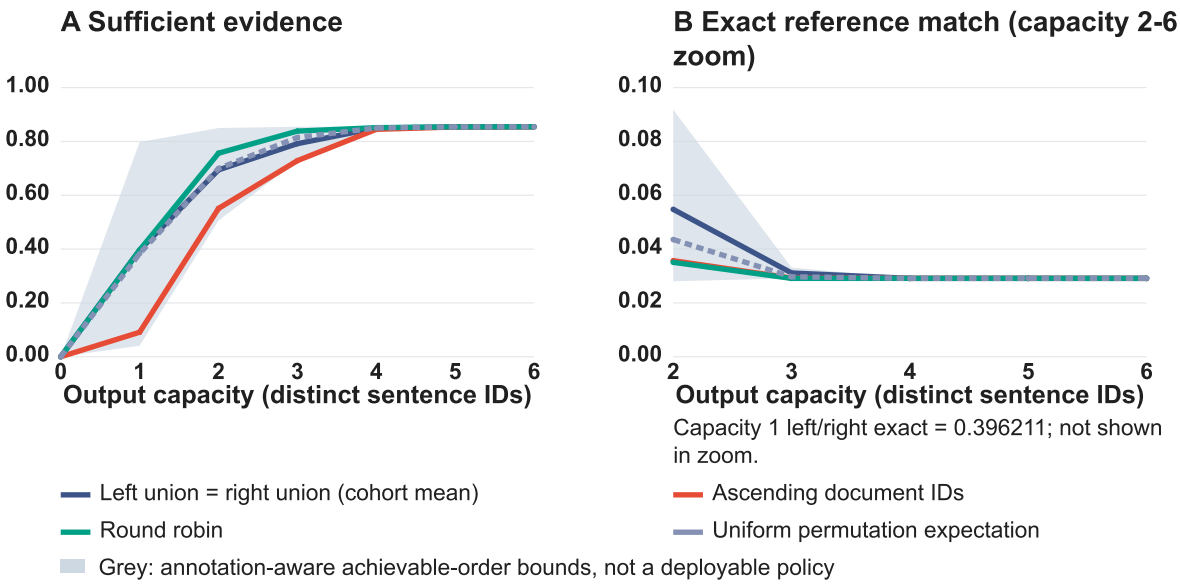
The cap-three sufficient score is 0.791700 for either source priority, 0.728582 for ascending IDs, 0.838328 for round robin and 0.814469 for exact uniform expectation. The annotation-aware lower and upper mean bounds are 0.727773 and 0.854553. Oracle knowledge must not be credited as a comparator's allowed information. The observed round-robin improvement is a finite property of this constructed bank, not a selected or independently tested agent intervention.

The endpoint tradeoff remains visible. At the same cap three, round robin lowers exact-reference mean from 0.031134 to 0.029119; the uniform expectation is 0.029697. A reordering that helps containment can hurt exact equality. The contained-reference sufficient probability is nondecreasing as capacity grows; exact-match probability need not be. For a four-ID union with one singleton rationale, the exact probability is one quarter at capacity one and zero at three. A scalar 'merge quality' claim would conceal that contract change.

A stronger universal-order claim is also false without annotations. Keep identical observed inputs A=[0], B=[1] and capacity one, but consider two otherwise admissible task worlds with reference [0] versus [1]. Any deterministic annotation-blind output is the same in both worlds and fails in one. The counterexample is a constructed information limitation, not a new SciFact instance or a frequency estimate. A method can exploit task text or learned scores, but then those additional rights and their costs must be declared and evaluated.

| Cap3 control | Sufficient mean | Exact mean | Information rights |
|---|---|---|---|
| Left / right priority | 0.791700 | 0.031134 | A,B only |
| Ascending IDs | 0.728582 | 0.029252 | A,B + document position |
| Round robin | 0.838328 | 0.029119 | A,B only |
| Uniform expectation | 0.814469 | 0.029697 | A,B; exact randomization average |
| Orbit lower bound | 0.727773 | 0.029119 | Auditor sees references |
| Orbit upper bound | 0.854553 | 0.033148 | Auditor sees references |

Table 22: Equal-task means on the same exposed bank; output capacity is three distinct sentence IDs. Uniform values are exact expectations, not Monte Carlo accuracy. Gold-aware bounds are not deployable baselines.




**C Every left-union failure receives one diagnosis (same pair denominator at every capacity)**

| Capacity | Pass | No rationale in union | Rationale cannot fit | Order contingent |
|---|---|---|---|---|
| 1 | 10,038 | 2,255 | 871 | 9,630 |
| 2 | 17,249 | 2,255 | 28 | 3,262 |
| 3 | 19,262 | 2,255 | 0 | 1,277 |
| 6 | 20,539 | 2,255 | 0 | 0 |

Cap3: 1,277 failures reverse successfully, yet reversing all pairs gives exact mean equality; real skill effect remains null.

Figure 7: New v16 capacity/order benchmark. Panel A shows equal-task sufficient means at capacities 0–6, Panel B exact means at capacities 2–6 with a 0.10 zoom, and Panel C a disjoint left-sufficient failure table. Grey regions are annotation-aware attainable-order bounds. Left and right priority coincide in the aggregate by cohort symmetry even though 1,277 individual failures reverse successfully. No curve is a natural skill-collapse rate or a token-matched model comparison.

### 32.3 Preserved v19: Benchmark alignment, reproducibility and the next behavioral gate

We reopened the official SciFact record and pinned evaluator source [1,2]. Its inspected abstract-level correctness checks document and label, then tests containment after slicing the supplied rationale list to three sentences. The local first-three endpoint conditions on correct labels and a supplied document; capacities above three and full exact-set equality are explicit diagnostic adaptations. We did not run retrieval, document selection, the native model baseline or official aggregate precision/recall. Comparison to a public SciFact system would require restoring those tasks and the exact split and scorer contract.

ERASER's original HTML separates agreement with human rationales from faithfulness to a model's prediction [4]. Our containment orbit studies the former kind of reference dependence; it does not measure how the evidence influenced an actual model. CheckList motivates focused behavior tests [5], but no CheckList package or language-model suite was run. We also reopened the pinned SkillsBench v1.1 release [19]: its native environment, oracle and verifier packages support a possible workflow-level follow-up. Reading its release is not an executed benchmark, and a skill/no-skill contrast alone does not supply independently qualified A/B composition.

Author-written verification avoids importing the analyzed scorer. Inclusion-exclusion over contained rationale events and direct alternative-size counting check 74,445 orbit probability cells. Full permutations for one union orbit of every observed size 0–6 check 6,111 prefix evaluations. A second direct

reconstruction checks 1,914,696 deterministic endpoint cells, all 22,794 pairs, all 147 aggregate rows and the 1,277 reversals. This is a distinct computation by the same author on the same exposed inputs. Nonauthor review, complete manuscript checks and any future empirical replication must be recorded separately.

The finite order benchmark does not add a single independent authoring history. Actual A/B histories and independently qualified pairs remain zero, and the qualified behavioral effect remains null. Plan, code and input locks make a development computation inspectable; they do not undo prior exposure or establish that a real skill merger follows any declared sentence operator. A behavioral claim still requires complete initiated histories, Q-only qualification without S feedback, fresh S arm outputs and fully charged comparison rights.

The next admissible behavioral benchmark should keep union, prefix3, round-robin and a simple randomized order as frozen controls on the same saved real arm outputs. It must specify document/ label rights, returned IDs and actual tokenizer costs, retain failures and all started authoring attempts, and use an independently reserved S cohort after Q-only qualification. An annotation-aware permutation envelope can then diagnose a measured loss without being mistaken for the deployed repair. A success criterion must be declared before the new S outcomes are exposed.

| Comparator | Current execution | Required boundary |
|---|---|---|
| SciFact [1,2] | Local conditional containment only | Native retrieval/labels/scorer not run |
| ERASER [4] | Method comparison; no model | Human agreement differs from faithfulness |
| CheckList [5] | Relationship-test design only | No package/model evaluation run |
| SkillsBench v1.1 [19] | Release reviewed; benchmark not run | Native environment; independent A/B/Q/S |
| Four order controls | 638,232 deterministic outputs | Exposed diagnostic IDs; no model/token match |
| Permutation envelope | Exact finite audit | Gold unavailable to deployed selector |

Table 23: Benchmark compatibility matrix. External literature descriptions are not local performance measurements; no external reported score is mixed with the present finite table.

## 33 Preserved v19: Agreement as a policy signal

V16 shows that a local input reversal can repair selected failures without improving the reversal-symmetric cohort. V17 asks a different operational question: can common evidence IDs supply a useful, label-blind composition signal? We keep the same 309 tasks, 204 recorded components, 2,494 inputs and 22,794 ordered pairs. Each constructor receives A and B only. Reference labels enter evaluation and training-component rule selection, never the output constructor. This remains an exposed diagnostic bank rather than independently authored skill behavior.

Let U be the distinct union, I its intersection and RR the deduplicated round-robin sequence. Agreement-first emits all common IDs, followed by RR noncommon IDs, then truncates to three. Agreement-only emits I without filling unused slots. Disagreement-first promotes the RR noncommon IDs before common ones and is declared a negative control. Left-union and RR retain their previous definitions. Fixed-count rules emit min(3,|U|) IDs; agreement-only emits |I|, possibly zero. Disagreement here means nonoverlap of IDs, not contradiction between scientific claims.

Agreement-only has a simple containment limitation. Because each A contains at most three distinct IDs, left-union truncated at three still contains all of A, while I is a subset of A. Any rationale contained in I is therefore also contained in left-union. Agreement-only cannot improve sufficient containment over left-union under this contract. Removing extra IDs can nevertheless improve exact-reference matching, so an exact-score increase need not indicate newly recovered information. Empty intersection is retained as a zero-score output, not omitted from coverage.

We compute two exact uniform-subset controls on the same U: length min(3,|U|), and length |I| for that particular pair. The second isolates whether the intersection selects unusually useful IDs after matching its shortened output count. These probabilities enumerate all equally likely subsets; they are not random model trials. ID-count matching does not equate tokens, inference access, bytes or elapsed cost. The conditional correct-label and supplied-document assumptions remain unchanged.

$$I = A \cap B \tag{41}$$

$$I \subseteq A \subseteq L_3(A, B) \tag{42}$$

$$S(I) \leq S(L_3(A, B)) \tag{43}$$

### 33.1 Preserved v19: Length-matched controls expose a reference-score tradeoff

The primary aggregate averages ordered pairs within each task, tasks within each recorded component, then the 204 components equally. Equal-task and pair-pooled alternatives remain separate. Of the 22,794 pairs, 5,407 contain a complete reference rationale in I, 15,132 contain one somewhere in

U but not I, and 2,255 contain none in U. Agreement-only passes sufficient containment on exactly the first stratum. Its 13,855 sufficient losses versus left-union coexist with 2,328 exact gains and 22 exact losses. This is a measurement tradeoff on constructed responses, not a behavioral skill success/ failure rate.

Agreement-only improves equal-component exact score from left-union 0.031228 to 0.150539 while reducing sufficient score from 0.789671 to 0.290889. It emits 20,604 IDs over all pairs, versus 58,284 for every fixed-count rule: pair-pooled average lengths 0.903922 and 2.556989 respectively. Against the pairwise length-matched uniform control, however, agreement-only is lower on both endpoints: sufficient 0.290889 versus 0.356638 and exact 0.150539 versus 0.221823. Thus its apparent exact gain against a longer rule cannot be credited to agreement information alone.

The bank includes 2,494 identical-input diagonal pairs and 20,300 distinct-input pairs; both sets still come from the same authored candidate lists. A distinct-input sensitivity keeps all 309 tasks and 204 components. Agreement-only remains below its matched-length random control: sufficient 0.240666 versus 0.311992 and exact 0.145307 versus 0.223723. Diagonal removal is a sensitivity, not a new independent cohort or a replacement primary denominator.

Agreement-first has 1,113 sufficient gains and 71 losses against left-union, but that bundled comparison also changes the noncommon ordering from left priority to RR. We inspected its first saved loss before writing a mechanistic claim and found that the common ID was already at the front. That example cannot identify the separate effect of promoting agreement. We therefore preserve the original primary run and add a separately planned, explicitly post-primary priority audit with a common RR baseline.

| Rule / control | Sufficient | Exact | Output length |
|---|---|---|---|
| left_union | 0.789671 | 0.031228 | min(3,\|U\|) |
| round_robin | 0.835115 | 0.029496 | min(3,\|U\|) |
| agreement_first | 0.835038 | 0.029496 | min(3,\|U\|) |
| agreement_only | 0.290889 | 0.150539 | \|I\| |
| disagreement_first | 0.842416 | 0.029496 | min(3,\|U\|) |
| uniform_cap | 0.811952 | 0.030012 | min(3,\|U\|) |
| uniform_common | 0.356638 | 0.221823 | \|I\| |
| oracle_pair | 0.840200 | 0.153696 | candidate-dependent <=3 |

Table 24: V17 equal-component scores on the same exposed bank. Uniform-common is exactly length-matched to agreement-only for every pair. The gold pair oracle ranges over left-union, RR, agreement-first and agreement-only only; it excludes the predeclared disagreement-first negative control and all random controls.

### 33.2 Preserved v19: A controlled agreement-priority counterexample

The auxiliary priority plan was written after the primary result and witness inspection. It reuses the same frozen pair records without changing the primary plan or candidate class. Against RR, agreement-first changes only the priority of common versus noncommon IDs: all common IDs fit within three, and the relative noncommon order stays fixed. This controlled finite intervention produces zero sufficient gains, two sufficient losses and 22,792 ties, with equal-component difference -1/13,056. Exact scores are tied on all 22,794 pairs. These observations are descriptive, not proof that agreement promotion must always lose.

The saved task scifact_v4_382 provides the actual counterexample. A=(1,2,0) and B=(4,5,0) each contain a valid alternative, either (1,2) or (4,5). Their common ID is 0. RR emits (1,4,2) and passes by containing (1,2); agreement-first emits (0,1,4) and contains neither complete alternative. The same union, capacity and output count are preserved, and the loss comes from promoting the shared item at the expense of completing either alternative. The two counted losses are ordered-pair records, not two independent tasks or naturally observed skill collapses.

The same controlled audit finds 122 sufficient gains and five losses when disagreement-first is compared with RR; its exact scores also tie throughout. Thus the negative control is not universally safe either. We do not retrospectively admit it to the four-rule selector or claim a deployable winner. Its role is to expose how the chosen candidate class and the constructed reference distribution affect an apparent consensus result. These counts do not establish that overlapping sentences are semantically wrong or nonoverlapping ones correct.

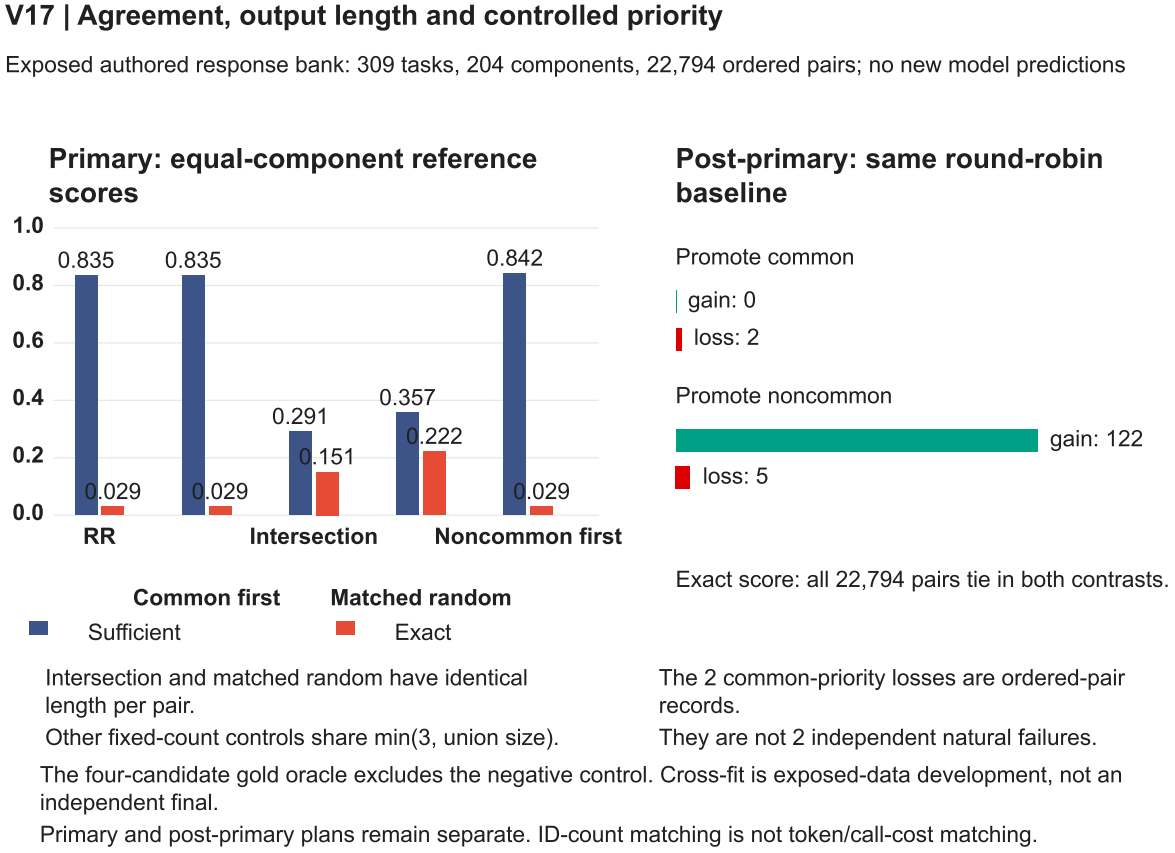



Figure 8: V17 agreement audit. Left: equal-component sufficient and exact scores, with agreement-only and uniform-common matched in output length pair by pair. Right: exploratory common/noncommon promotion against the same RR baseline, retaining gains and losses. Both panels reuse the exposed authored bank; the right panel follows a separately recorded post-primary plan. No model predictions, independent histories or population confidence intervals are added.

### 33.3 Preserved v19: Grouped rule selection does not repair the selection boundary

To keep a gold-aware chooser separate from a possible rule selector, the primary plan assigns each recorded component to fold int(component_SHA256[:16],16) mod 5. The five held-out component counts are 40, 38, 38, 42 and 46. For each fold, only the other components may choose among the four declared candidates: left-union, RR, agreement-first and agreement-only. Training criterion is the same equal-component mean, with fixed declaration-order ties. Sufficient and exact are separate objectives; no held-out component contributes to that fold's choice.

All five sufficient-objective folds choose RR; all five exact-objective folds choose agreement-only. Cross-fitted equal-component scores therefore equal those fixed rules: sufficient-objective score 0.835115 and exact-objective score 0.150539, with no added score from varying the selected rule. The cross-objective outcomes remain visible: the exact-selected rule has sufficient 0.290889, while the sufficient-selected rule has exact 0.029496. This stable-choice null result is retained rather than describing the fold procedure itself as an improvement.

The per-pair gold oracle over the same four candidates reaches 0.840200 sufficient and 0.153696 exact, exceeding the respective cross-fitted scores by 0.005085 and 0.003158. It uses the held-out reference to choose independently for each pair and endpoint, so it is a diagnostic class upper bound, not a deployable routing score. The predeclared disagreement-first negative control, excluded from this class, scores sufficient 0.842416 and can exceed that four-candidate oracle. There is no contradiction: a restricted-class maximum does not bound methods outside its class. Adding that method after seeing its result would be another selection step, not a validated improvement.

Component-disjoint folds prevent the same recorded component from appearing in both training and held-out roles for a particular decision. They do not erase the earlier exposure of all annotations and diagnostic designs. These are finite development reanalyses, not a secret final evaluation, independent model histories or a population estimate. We report exact fractions and paired changes without treating 22,794 constructed pairs or ten selection decisions as independent observations.

### 33.4 Preserved v19: Agreement literature, fair comparisons and the next admission gate

Self-consistency [20] samples multiple reasoning paths and aggregates their final answers. Our intersection signal concerns sentence IDs in two fixed constructed evidence lists; no reasoning-path sampling, majority answer vote or native self-consistency benchmark is implemented. The agreement question is relevant, but importing that method's published accuracy or treating sentence overlap as its mechanism would conflate different information contracts.

Cawley and Talbot [21] discuss overfitting a finite-data selection criterion and resulting evaluation bias. We reviewed the official abstract rather than transferring a theorem. The component-held-out rule chooser makes local selection rights explicit but cannot retrospectively make an exposed diagnostic bank independent. Prior SciFact, ERASER and SkillsBench distinctions remain: supplied-reference agreement is not model faithfulness, correct-label component scoring is not native end-to-end verification, and an unrun native skill benchmark is not qualified A/B composition evidence.

Both the primary audit and post-primary priority audit freeze plans, code, alternative verifiers and input bytes before their own output generation. Separate author checks reconstruct all 113,970 deterministic outputs, 25,119 uniform probability cells, 48 aggregate rows and ten fold choices; the auxiliary verifier reaggregates four controlled contrasts from 22,794 saved pairs. These are alternate implementations by the same author, not external peer review. Actual independent A/B authoring histories and qualified pairs remain zero; the target behavioral composition effect remains null.

The next study must freeze the candidate class, exact output/label contract and actual token/call costs before collecting independently authored, Q-qualified A/B responses on a reserved S cohort. It should test simple RR, agreement promotion, intersection and predeclared negative controls with all failures and empty eligibility retained. A successful correction on this exposed construction cannot substitute for that evidence. The immediate recommendation is a stronger comparison contract, not deployment of the highest post-hoc score.

### 33.5 Preserved v19: Capacity windows separate order effects from scorer truncation

A new predeclared finite audit keeps the round-robin relative order within common and noncommon IDs and only promotes the common class. For a nested output prefix L, the first sufficient capacity is the minimum, over complete reference alternatives, of the largest reference-ID rank. Two operators therefore differ in sufficiency on at most one contiguous capacity interval for a fixed pair. This is an annotation-containment identity, not a semantic claim about agreeing natural skills.

We evaluated 22,794 existing ordered pairs at capacities0..6, retaining309 tasks and204 components. Full-output sufficient gain/loss counts are565/2700 atcap1,95/25 atcap2,0/2 atcap3 and0/0 atcap4..6. The first-three endpoint still has0/2 atcap4..6: a larger generated output does not remove a grader's first-three restriction. Capacities count output IDs; token and invocation costs remain unmeasured.

In the declared <=3-ID input class, if both inputs are individually annotation-sufficient, stable common promotion cannot lose full-output sufficiency at capacity>=4. An empty intersection leaves the baseline unchanged; it does not imply that both outputs succeed. Exhaustive authored certificates cover3,046 input pairs and122,571 one/two-alternative reference families. They verify the stated finite boundary and include overlapping alternatives, rather than adding new independent observations.

$$r(L) = \min_{R \in \mathcal{R}} \max_{e \in R} \mathrm{rank}_L(e) \tag{44}$$

$$U_k(L) = \mathbb{1}[k \geq r(L)] \tag{45}$$

| Output capacity | Full gain | Full loss | First-three gain/loss |
|---|---|---|---|
| 1 | 565 | 2700 | 565 /2700 |
| 2 | 95 | 25 | 95 /25 |
| 3 | 0 | 2 | 0 /2 |
| 4..6 | 0 | 0 | 0 /2 |

Table 25: Exposed finite controls; full containment and first-three are distinct endpoints.

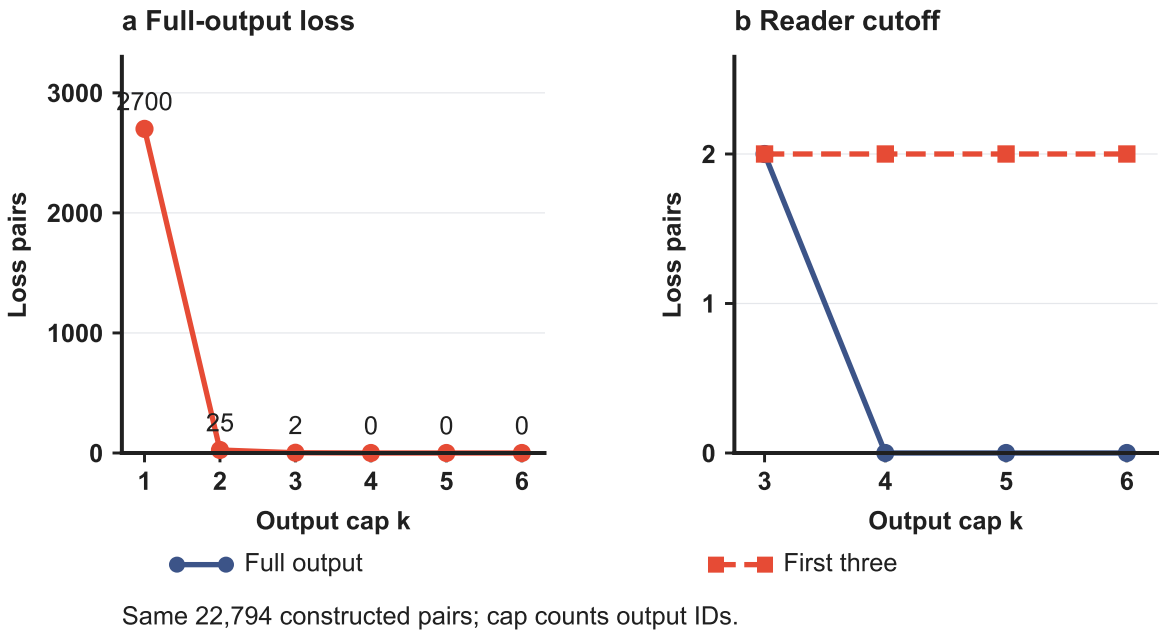



Figure 9: New v19 finite containment audit on the same22,794 constructed ordered pairs. Left: full-output loss counts across capacities. Right: cap3..6 zoom distinguishes recovery in full containment from the unchanged first-three loss. Neither curve is a real-model or independently qualified skill benchmark.

### 33.6 Preserved v19: Weighting and information rights limit a universal merge claim

The both-input annotation-sufficient stratum contains12,067 ordered pairs from308 tasks and203 components. Atcap1 its pooled contrast is-844/12067 while its equal-component contrast is+3363459013/266277211200. This predeclared descriptive sensitivity changes sign because response multiplicities weight units differently. We do not choose the favorable weighting after seeing the result or treat repeated pair views as new samples.

A concrete preserved failure has A=[1,2,0], B=[4,5,0] and alternative references{1,2} or{4,5}. Round-robin cap3=[1,4,2] contains a complete reference; promoting common ID0 gives[0,1,4] and loses it. Atcap4 full containment recovers but first-three does not. If same-length output sets differ, a singleton reference from the removed IDs also prevents a reference-blind universal no-regression guarantee. Such a constructed annotation witness is not evidence of how often real model skills conflict.

The original author verifier mistakenly strengthened no loss into universal success for disjoint inputs. Its failed frozen execution is retained. A separately frozen corrected verifier and verification history check the proper statement and the original unchanged analysis. Independent authoring, Q qualification, reserved S evaluation and measured token/call costs remain absent; the behavioral composition effect stays null.

## 34 Preserved v19: Discussion and Limitations

The strongest warranted conclusion is a prerequisite decision: admit the measurement contract before interpreting a qualified-composition contrast. The audit demonstrates particular failure modes in an incomplete grader and particular order sensitivity in a ranked endpoint. It does not demonstrate that combining real skill files generally lowers completion.

### 34.1 Preserved v19: A behavioral study that could falsify the hypothesis

A next study must justify new source/history separation and access controls rather than rename the exposed tasks. Freeze the ancestor, resource graph, author contexts, proposal cap, Q gain rule, final endpoint, and AB order before S. Retain failed proposals and nonqualifiers. Use Q only for qualification, not retrospective repair based on S. If no pair qualifies, report yield zero and effect null; weakening the gate after seeing joint failure would redefine the target.

For eligible pairs, allocate fresh matched S0/A/B/AB resets. Keep all assigned availability cells, including nonactivation, invalid output, agent-caused timeout, and deliberate deferral. Unresolved external outages require a separate missingness ledger and bounded success summaries. Record installed, routed, read, and assembled hashes. Diagnostics should include BA order, tokenizer-matched

irrelevant context, structured padding, unchanged-resource controls, and sham repair. A negative required-ranking score remains operational loss even if semantic-loss attribution is falsified.

Repeated trials share histories and source families. An inferential plan must say whether histories and families are fixed or sampled, and model both dependencies for claims about both populations. Qualification yield and paired variability from an independent feasibility pilot should inform later precision requirements; zero current histories cannot support a power claim. The semantic-interference explanation is weakened if the negative contrast vanishes when full evidence, ordering, tokenizer length, rights, and the declared grader are controlled.

Admit task rights, answer expressibility, annotation alternatives, grader behavior, and reset/cost contract.
Freeze S0 and independent author contexts; generate capped A/B proposals using D only.
Preserve all proposals and failures; apply the predeclared auditor-only Q gate to both branches.
If eligible count is zero, return yield zero and null composition effects.
Freeze eligible candidate/resource bytes and joint order; assign fresh S0/A/B/AB resets and controls.
Compare policies under equal feedback rights and full budget; retain deferral, missingness, and nonactivation.
Estimate the mean-isolated-branch contrast with declared family/history weighting.
Attribute mechanisms only to the extent supported by matched interventions.

Algorithm 6: Prospective protocol. The current deterministic audit completes admission checks only; it does not execute the authoring-to-S sequence.

### 34.2 Preserved v19: Strong comparisons and accounting

A fixed-pool adapter comparison would give every policy identical candidate bytes, development feedback, tool rights, and total budget. Conservative replay must be allowed to narrow, reject, or fall back; an always-merge straw controller is insufficient. Typed migration must preserve the task contract and pass behavior checks, not receive credit for schema legality alone. Assessment-based assembly must retain router/assembler costs and information access. Naive concatenation is a separate control, not native SkillCAT v2 [16].

A native-system comparison is a different experiment: generation procedures, teacher/reference access, training objectives, routing, and assembly all matter. We have executed neither that experiment nor the fixed-pool adapters. Candidate-conditioned stress sets [17] can diagnose a policy but cannot estimate unselected prevalence. Instruction/content/resource controls [18] and versioned workflow packages [19] are useful extensions only after their own rights, environments, and contracts are frozen.

Charge candidate creation, rejected proposals and repairs, replay, routing, assembly, loading/cache, retries, and final execution. Q/S research overhead can be reported separately but remains part of total resource use. Compare completion alongside deferral, conditional deployment coverage, and cost; a policy that avoids every hard deployment must not appear unconditionally superior. No native-paper score is placed beside the local diagnostic counts as if measured on one benchmark.

| Comparison mechanism | Allowed conservative action | Essential fairness condition |
|---|---|---|
| Source replay / SkillCommit-inspired | Reject, narrow, split, or fall back | Same source feedback and charged replay budget |
| Typed migration / SkillSmith-inspired | Repair callers and interfaces | Same resources; behavioral acceptance beyond static legality |
| Assessment/assembly / SkillCAT-inspired | Select and assemble runtime content | Charge LLM routing/assembly; preserve information rights |
| Naive joint availability | Canonical unmodified installation | Same ancestor, reset, tools, endpoint, and execution budget |

Table 26: Prospective comparison mechanisms. An inspired adapter is labeled as such and never reported as a native reproduction.

### 34.3 Preserved v19: Threats to interpretation

Gold labels and rationale sets are privileged inputs. Their semantic correctness and completeness were not independently adjudicated. A sentence outside the annotation union is unannotated by construction; calling it irrelevant is operational shorthand, not a domain-expert judgment. Exact-annotated can reject valid unlisted evidence, and sufficient containment tolerates unhelpful extras. Our conclusions concern the specified annotation contract.

The study uses one exposed scientific-verification adaptation. Structural disjointness does not establish semantic independence or absence of model pretraining exposure. Four paired flips and one unreachable case are finite findings worth preserving, not a population prevalence claim. The related-work inspection is bounded to recorded primary passages; several references are preprints, one uses an abstract-only scope, and none supplies a reproduced external result. No external professor or independent scientific reviewer endorsed the results.

The elementary reachability proof and counterexample do not certify methodological novelty. We do not show a new integration controller, general harm from skills, or superiority over the cited systems. The falsifiable contribution is narrower: concrete admissibility checks and controls prevent unsupported inferences in the planned qualified-composition experiment. Full reference scopes, local provenance, and reproducibility limits are preserved in the appendices and companion records.

The new support certificate is complete only under the declared ordered-ID response contract and its checked outside-union condition. It does not adjudicate unannotated valid rationales or model-generated labels. Its sharpness permits arm/task-specific response choices, without common-program, routing, qualification or learning constraints. These deliberately weak behavioral restrictions make the envelope conservative for actual skills and unsuitable as an empirical rate. Changing benchmark annotations, response validity, weighting or Q selection requires a new certificate or transport analysis. The total-variation bound additionally requires common preselection weights and fixed candidates; it does not absorb authoring changes or final-data feedback.

## 35 Preserved v19: Conclusion

Before asking whether useful skills fail together, specify which outcomes would count and whether valid answers can be expressed. On the preserved SciFact adaptation, original alternatives, a three-ID cap, and evidence ordering change diagnostic acceptance in quantitatively different ways. These results support grader-admission decisions and narrower mechanism claims. Qualified composition remains a separate empirical question: with zero actual histories, its effect is null. The next defensible advance is an independently qualified, freshly evaluated behavioral study with strong conservative comparisons and complete accounting.

The new v14 analysis closes a narrower gap: it computes exact response-support envelopes and distinguishes their fixed-cohort meaning from qualification-cohort transport. The observed zero envelope at budgets at most three and the large budget-four worst case change how a future score sign should be audited. They do not supply missing qualified histories. The next decisive evidence remains an independently authored, fully charged A/B-to-S experiment with an unchanged declared task contract.

The constrained v15 replay adds a concrete falsification control: a sufficient-score gain and an exact-score loss coexist under a sentence-preserving union, while a three-ID cap introduces real omission. This is a measurement-and-operator result. It narrows what a future negative behavioral score would mean without supplying the still absent qualified histories.

The v16 orbit benchmark makes one repair claim precise: reversing priority fixes all selected cap-three losses but cannot improve a reversal-symmetric cohort. Matched simple controls and an exhaustive reference-conditional failure partition locate a future intervention's burden of proof. Round-robin containment gains with exact-match losses remain finite development evidence; qualified behavioral effects still await independent histories.

V17 shows that common-ID agreement, shortening and model selection require separate controls. A same-RR priority intervention has two actual sufficient failures; intersection-only trades coverage for exact agreement and still trails a length-matched random subset. Component cross-fitting adds no score over its stable fixed choices. The oracle is restricted to four predeclared rules, while excluded negative-control results remain diagnostic. None of these finite findings identifies a qualified behavioral skill-composition effect.

V19 identifies the capacity mechanism more precisely. Completing an alternative reference set is controlled by its maximum output rank; a stable common-item promotion creates at most one contiguous gain or loss window. For two individually annotation-sufficient inputs of length at most three, full-sufficiency loss is impossible at cap four or above. First-three scoring can nevertheless retain the earlier losses, and cap-one pooled and equal-component contrasts have opposite signs. These exact conditional statements refine the finite mechanism claim; they do not establish a behavioral effect of independently authored executable skills.

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## A Reproducibility and preserved artifacts

The v11 audit is runs/research_progress_v11_20261018_01/skillmerge/audit_01. The v12 project is projects/research_manuscripts_v12_20261018/skillmerge; its executed output is runs/research_manuscripts_v12_20261018_01/skillmerge/measurement_01. The output preserves locked_plan.json, results.json, task_measurements.json, diagnostic_rows.jsonl, the exact consumed measure.py snapshot, and receipt.json. The project preserves verify.py and VALIDATION.json. The v13 manuscript does not overwrite those files, earlier manuscripts, or active skills.

Reproduce using Python 3.11: python -X utf8 -B projects/research_manuscripts_v12_20261018/skillmerge/measure.py –out runs/research_manuscripts_v12_20261018_01/skillmerge/measurement_fresh. Choose a genuinely new child if that path exists. The –self-check option runs finite mathematical/software checks without an agent. verify.py documents saved-run reaggregation. No new dependency or model API is required.

The v12 receipt SHA-256 is 93e6014be9c1b980856c95194b3c0e8edb7e0f758fd233ef45c867ab3517bae3. Its original binding omitted the consumed prepared manifest; VALIDATION.json explicitly supplements that omission against the pre-existing sealed v11 receipt and retains the original receipt unchanged. This is a documented integrity repair, not a claim that the historical receipt originally included every input. Hashes bind local bytes, not semantics, publisher authenticity, or independent replication.

## B Endpoint and finite-control details

The unique four-sentence-only task is scifact_v4_264 with rationale IDs (7,8,9,10). It is retained in both denominator and source records. The pinned evaluator's metrics.py lines 10–52 defines the three-ID rationalized-abstract rule; data.py lines 218–242 preserves the supplied response list in the inspected parser. Our document-prefix construction chooses ascending document order; upstream parser sorting is not asserted.

The annotation-reachable counts for budgets 1–10 are (290,305,308,309,309,309,309,309,309,309). Constructed prefix sufficient passes are (17,43,76,119,148,176,202,221,238,251); first-three passes are (17,43,76,76,76,76,76,76,76,76); exact-annotated passes are (17,3,0,0,0,0,0,0,0,0). Every count uses 309 evidence tasks. The 222 insufficient tasks are not silently mixed into this positive-task figure.

The equal-task uniform-subset containment expectations for budgets 1–10 are (0.190907,0.352965,0.490307,0.606539,0.701456,0.776807,0.834577,0.877959,0.910136,0.935464). They condition on auditor-correct labels and size min(b,n). Distinct rationale sets are deduplicated before inclusion–exclusion; overlapping unions are supported. Enumerating all subsets of each relevant size independently checks the formula. The displayed decimals are rounded summaries, not random-trial estimates.

In the earlier grader audit, 770 original annotation-or-empty responses were accepted by the alternative-preserving local contract; 531 wrong-label, 592 one-sentence-deleted, and 6,903 malformed-response diagnostics were rejected. These groups are different constructions and denominators, not a single overall model accuracy. The current budget/order study preserves those earlier findings without merging them into a behavioral performance table.

## C Source review and claim boundaries

SOURCES.json links the 19 prior references and two v17 additions to inspected primary URLs, selected passages, intended claims, exclusions, and acquisition receipts. Conference records establish bibliographic identity where available; pinned arXiv revisions establish the preprint identity actually inspected. For Underspecification the review scope is the official JMLR metadata and abstract, not the entire paper. Other full HTML pages were inspected at the recorded relevant sections; availability of HTML does not mean every appendix was audited.

A separate standard-library retrieval records exact HTML/plain-code response bytes, UTC acquisition time, content type, and SHA-256 without storing whole copyrighted papers. These response hashes do not identify PDF bytes or the browser's rendered representation. Failed guessed locators are explicitly rejected rather than used to complete metadata. The five recent skill preprints resolve to matching primary versions; their dates alone are not grounds to fabricate or discard bibliographic facts.

CLAIMS.json separates local measurements, author-derived finite statements, literature descriptions, proposed protocols, and untested hypotheses. Figure plans bind each plotted value to result keys. Korean explanations provide the derivations, objections, source differences, and independent-study

design. Neither this documentation chain nor same-session cross-review constitutes external scientific validation.

## D Completeness and sharpness of the new response-support certificate

Let the annotation family be nonempty and contain no empty set; let d be an ID outside its union. For any response L, T(L) and E(L) imply U(L). The empty response witnesses (0,0,0) at cost zero. If E(L)=1, its full set equals an annotation R. Enumerating each min(3,|R|)-ID prefix subset of R enumerates every possible T score for that exact set, so the recorded minimum length is complete for (1,0,1) and (1,1,1).

If U=T=1 and E=0, the scored prefix contains an annotation R of size at most three. Any such nonexact response has length at least |R|+1. The response R followed by d has exactly that length, keeps U=T=1, and cannot equal any original annotation because d occurs in none. This proves both attainability and minimum length for (1,1,0).

If U=1,T=0,E=0, L has at least four IDs. Its three-ID prefix P contains no annotation, and some annotation R lies in L. Let W=P union R. If W is not an annotation, P followed by R minus P is a witness no longer than L. If W is an annotation, L must have at least one further ID because L is not exact. Appending d to the reduced witness therefore still fits within |L|. The prefix remains unsuccessful and the outside ID guarantees nonexactness. Exhausting P and R is consequently complete and retains a minimum-length (1,0,0) witness. These five cases exhaust all possible patterns. The outside-union assumption is necessary for this particular reduction argument; its failure must be detected rather than hidden.

For discrepancy e_t(L)=g(L)-h(L), the change in a three-arm mean-isolated contrast is e_t(L_AB) minus half the sum of e_t(L_A) and e_t(L_B). This lies in the interval from m_t-M_t to M_t-m_t. Finite support attains both endpoints by choosing minimizing/maximizing witnesses. Because no cross-task or cross-arm realizability restriction is imposed, nonnegative fixed weights preserve sharpness after summation. Imposing a shared learned program or information constraints can shrink the attainable set, so the certificate remains an outer bound rather than a claim that qualified agents attain it.

For two selected distributions derived from the same original weights, their common probability mass after normalization is r/max(q_g,q_h). Total variation equals one minus that overlap. A bounded function in [-1,1] has expectation difference at most twice total variation, attained by assigning +1 and -1 according to the sign of the distribution difference. An authored three-history example checks equality. If an admission mass is zero, normalization is undefined and no point effect or overlap correction is supplied.

## E New execution and a bounded candidate-history pilot contract

The canonical new result is runs/research_advancement_v14_20261019_01/skillmerge/bounds_01/results.json, SHA-256 f70e47a5b1455cfe27a85f8dcb6ac31e11a205221519aa5f99a030173b74d106. The adjacent receipt binds the locked plan, code and input snapshots. The validation_01 sibling preserves a separate verifier snapshot, input lock, result and receipt. The published source manifest records exact paths and byte hashes; these are reproducibility bindings, not proof of annotation truth or external replication.

A proposed next feasibility pilot uses three independently sourced task families and four independent authoring attempts per family: twelve attempted A/B histories, not a power-calculated confirmatory sample. The two authors receive the common ancestor, public D information and identical tool rights, but not each other's patch or auditor-only Q/S labels. Cap each branch at two proposals and preserve every rejection, invalid proposal and failed call. Before authoring, freeze the actual model identity, request/token/cost limits, rationale contract and tokenizer-aware budget. Use twenty source-disjoint Q tasks per family and one predeclared seed; require each isolated branch to exceed the same ancestor by at least two successful tasks out of twenty. Freeze all capped proposals before Q and return no Q feedback to their authors. An auditor chooses the first passing proposal in the predeclared proposal order for each branch, not the best final-S result. With one ancestor and at most four branch proposals, qualification costs at most one hundred task-arm executions per history, before charged generation and tool work. This is a fixed feasibility rule, not a powered significance test, and permits no final-S repair. The number of eligible pairs is an outcome: do not relax the gate if it is zero.

For each eligible pair, a bounded first S phase would assign four fresh reset arms on twenty unexposed tasks per family with two predeclared seeds per arm, up to 160 final task-arm-seed executions per pair. Twenty tasks and two seeds are feasibility caps, not an asserted precision or independence

guarantee. A third party or user must provide genuinely unexposed source histories/tasks, and the evaluation package must preserve their access log. Record all four frozen outputs before endpoint rescoring, a union-of-Q-gates cohort ledger if testing qualification changes, complete availability/routing/read/assembly traces, rejected proposals and total resource use. This protocol is unexecuted and requires separately authorized model/data-transmission limits; the current offline run initiates none of those calls.

## F V15 executed replay inventory

The current structured manuscript, guide, PLAN.json, analyze.py and independent verify.py are in projects/research_advancement_v15_20261019/skillmerge. The executed snapshot is runs/research_advancement_v15_20261019_01/skillmerge/analysis_01, containing exact consumed code/plan, input_lock.json, immutable input copies, task_results.json, results.json and receipt.json. Earlier version paths elsewhere in this paper identify historical computations, not the present manuscript source.

Reproduction requires Python standard library only and a new output directory: python projects/research_advancement_v15_20261019/skillmerge/analyze.py –out runs/research_advancement_v15_20261019_01/skillmerge/NEW_RUN. Read-only verification is python projects/research_advancement_v15_20261019/skillmerge/verify.py. The replay generated45588 deterministic operator applications; real skill histories and model API calls remain0.

## G V16 frozen order/capacity benchmark inventory

New artifacts reside under runs/research_advancement_v16_20261019_01/skillmerge/analysis_01. PLAN.json, analyze.py, verify.py and all three source input snapshots are copied before analysis, and input_lock.json records hashes, environment and UTC time. orbit_results.json stores 3,545 distinct task/union orbits with seven capacities, task_results.json stores 2,163 task/capacity summaries, and results.json stores 147 method/capacity/endpoint aggregates plus the complete failure partition. Witnesses and the receipt preserve counterexamples and output bindings.

Use the frozen analysis script only with a new output directory; the default code rejects any existing target. The verifier is read-only. Inclusion-exclusion accounts for overlap among supplied alternative rationales rather than adding their probabilities as if disjoint. Uniform permutation expectation assigns equal mass to size-k subsets because each has k!(n-k)! full-permutation completions. This elementary identity also explains why no sampling error or seed confidence interval is reported.

Figure 6 is generated from saved exact aggregate fractions by make_figure.py using standard-library SVG only. Its receipt records all 72 plotted values and input/output hashes; existing figures are preserved. Primary-source review notes store only scoped paraphrases and URLs; no arbitrary PDF or dataset was downloaded. All numeric findings refer to the local exposed fixture, never an imported literature score.

## H V17 agreement and post-primary priority inventory

analysis_01 freezes the primary PLAN, analyze.py, verify.py and three existing inputs; pair_rows.jsonl retains 22,794 ordered records and fits the existing 8 MB text-view limit. task_results.json preserves 309 task summaries, selection.json ten fold decisions and witnesses.json actual improvement/regression examples. The primary verifier uses separately written constructors and inclusion-exclusion rather than importing the author functions.

priority_audit_01 is separately registered after primary-result/witness inspection and preserves the original primary plan and outputs. Its PLAN states that left-versus-agreement bundled comparisons cannot isolate common priority. It freezes the saved primary pair rows and records controlled RR differences, witnesses and a separate reaggregation verifier. Neither analysis creates independent skill histories, new predictions or natural sample frequencies.

## I Source inspection boundaries

[1] ACL bibliographic record and abstract; arXiv v2 Sections 3.3 and 4.1–4.2. Original rationale alternatives, abstract-level containment/cap, and separate sentence-level precision/recall. v2's statement that the longest rationale has three sentences is not transferred to the later local raw snapshot, which contains one four-sentence-only task.

[2] metrics.py lines 10–52: MAX_ABSTRACT_SENTS=3 and supplied-list slicing before containment. data.py lines 218–242: sentences assigned to prediction rationale without sorting in the inspected parser. Sentence endpoint is separate. Local document-prefix ordering is an authored diagnostic policy, not parser-imposed sorting. V16 reopened the pinned metrics.py source, especially

MAX_ABSTRACT_SENTS and is_correct; locally reimplemented conditional scoring, not official end-to-end evaluation.
[3] ACL identity/abstract and original HTML evaluation discussion, including label-plus-evidence versus label-only scoring and Section 5.2 warning that evidence annotations may be incomplete. No reported baseline percentage is imported.
[4] ACL identity/abstract and HTML Sections 1 and 4. Agreement with human rationales and faithfulness to the model's prediction are distinct. Its confidence-perturbation sufficiency is not our annotation-set sufficient-containment endpoint. V16 reopened arXiv HTML v2 Sections 1 and 4.1–4.2; plausibility/faithfulness distinction rechecked, no model or official benchmark result reproduced.
[5] ACL identity/abstract and HTML Section 2.2 minimum functionality, invariance, and directional expectations. Supports focused behavioral/relationship tests; current tests exercise deterministic graders rather than a model through CheckList software.
[6] Pinned v2 identity and Section 1.1 multi-metric measurement/targeted evaluation/standardization. Supports reporting different task desiderata separately. No external model numbers are copied.
[7] Pinned v3 title/lead author and Sections 3.4.2–3.5. Metric/task specification affects observed capability patterns. Cite the inspected arXiv revision, without guessing a journal publication year.
[8] Title/authors and Sections 2–3. Score-to-scientific-inference validity depends on intended inference, assumptions, and supporting/refuting evidence. The pinned title is used; later differently titled publisher metadata is not merged.
[9] Identity and selected Sections 3.2–3.4 and 3.8. Auxiliary formatting demands, reuse limitations, and phenomenon-to-task-to-code-to-claim links. No survey percentage is imported.
[10] Identity and Sections 3–4/Appendix E.1. Declarative modules and compilation; a specified metric filters traces. This is the inspected v1 title, not a merged later conference-title record.
[11] Official JMLR metadata and abstract only. Similar held-out predictive performance can coexist with different deployment behavior. Full article not claimed as read.
[12] Pinned v8 identity/abstract and documentation categories covering motivation, composition, collection, and recommended use. Supports explicit source/split/exposure documentation; no certification of dataset quality.
[13] Title/authors, abstract, and discussion of benchmark/task choice affecting apparent method rankings. Supports fixing weighting and task family coverage before final comparison; not a local ranking result.
[14] Primary identity and selected patch/replay/compatibility/consolidation sections. Source-case replay and narrowing/rejection/splitting before commit. Teacher and executor information access differ. No numerical result is imported.
[15] Identity, selected typed operations in Section 3.4 and appendix interaction assumptions. Patch/Retype/Extract/Inline and required caller updates. Static legality is not behavioral qualification; its interaction assumptions are not theta_Q.
[16] Identity and selected test-time routing/assembly section, equations 5–6. v2 uses an LLM router and an LLM assembler over selected topology nodes. Concatenation is a separate control, not labeled native SkillCAT.
[17] Identity and selected CCV/counterfactual validation and Section 4 passages. Candidate-conditioned regression validation/repair; the source itself distinguishes its operational analysis from formal causal identification under stochastic or multi-edit settings.
[18] Identity and Section 3 controls C0–C3. SKILL.md injection versus separately mounted assets, target/irrelevant byte-matched instructions, and slices. Byte matching is not equality under the actual model tokenizer; reported performance not imported.
[19] Official release sections describing task/environment/oracle/verifier packages and pinned 0.6.x tooling. Prospective workflow benchmark requires its native environment and skill/no-skill contract. Release inventory is not reproduced as a local result. V16 reopened official v1.1 release's native package and with-skill/no-skill instructions; release artifact or task execution not downloaded or run.
[20] Primary HTML Section 2 and nearby introduction reviewed: sampled reasoning paths and aggregation of final answers. Our sentence-ID intersection is neither a reproduction nor a proxy performance result; no external score imported.
[21] Official metadata and abstract only. Finite-data selection can overfit; no full-text theorem or model-selection experiment reproduced, and component folds do not restore independent final data.